

Deploy and configure Azure Firewall using the Azure portal

Controlling outbound network access is an important part of an overall network security plan. For example, you might want to limit access to web sites. Or, you might want to limit the outbound IP addresses and ports that can be accessed.

One way you can control outbound network access from an Azure subnet is with Azure Firewall. With Azure Firewall, you can configure:

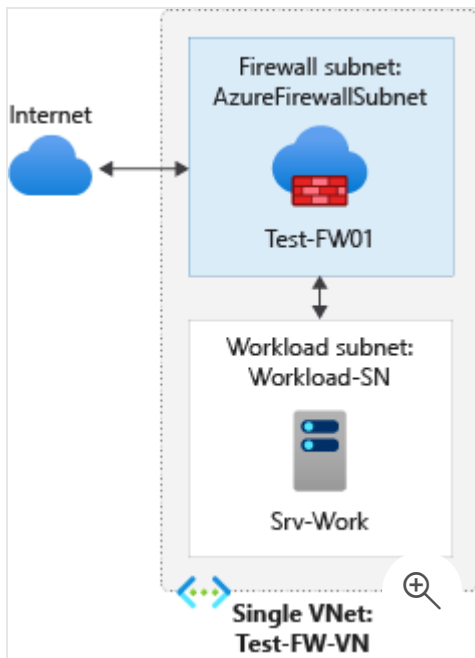
- Application rules that define fully qualified domain names (FQDNs) that can be accessed from a subnet.
- Network rules that define source address, protocol, destination port, and destination address.

Network traffic is subjected to the configured firewall rules when you route your network traffic to the firewall as the subnet default gateway.

For this article, you create a simplified single virtual network with two subnets for easy deployment.

For production deployments, a [hub and spoke model](#) is recommended, where the firewall is in its own virtual network. The workload servers are in peered virtual networks in West US with one or more subnets.

- **AzureFirewallSubnet** - the firewall is in this subnet.
- **Workload-SN** - the workload server is in this subnet. This subnet's network traffic goes through the firewall.



In this article, you learn how to:

- ✓ Set up a test network environment
- ✓ Deploy a firewall
- ✓ Create a default route
- ✓ Configure an application rule to allow access to www.google.com
- ✓ Configure a network rule to allow access to external DNS servers
- ✓ Deploy Azure Bastion for secure VM access
- ✓ Test the firewall

ⓘ Note

This article uses classic Firewall rules to manage the firewall. The preferred method is to use [Firewall Policy](#). To complete this procedure using Firewall Policy, see [Tutorial: Deploy and configure Azure Firewall and policy using the Azure portal](#)

If you prefer, you can complete this procedure using [Azure PowerShell](#).

Prerequisites

If you don't have an Azure subscription, create a [free account](#) before you begin.

Set up the network

First, create a resource group to contain the resources needed to deploy the firewall. Then create a virtual network, subnets, and a test server.

Create a resource group

The resource group contains all the resources used in this procedure.

1. Sign in to the [Azure portal](#) .
2. On the Azure portal menu, select **Resource groups** or search for and select *Resource groups* from any page. Then select **Create**.
3. For **Subscription**, select your subscription.
4. For **Resource group** name, type **Test-FW-RG**.
5. For **Region**, select **West US**. All other resources that you create must be in West US.
6. Select **Review + create**.
7. Select **Create**.

Create a virtual network

This virtual network has two subnets.

ⓘ Note

The size of the AzureFirewallSubnet subnet is /26. For more information about the subnet size, see [Azure Firewall FAQ](#).

1. On the Azure portal menu or from the **Home** page, search for and select **Virtual networks**.
2. Select **Create**.
3. On the **Basics** tab, configure the following settings:

 Expand table

Setting	Value
Subscription	Select your subscription
Resource group	Test-FW-RG
Virtual network name	Test-FW-VN
Region	West US

4. Select **Next**.

5. On the **Security** tab, configure the following settings:

 Expand table

Setting	Value
Enable Azure Firewall	Selected
Azure Firewall name	Test-FW01
Tier	Standard
Policy	None (Use classic firewall rules)
Azure Firewall public IP address	Select Create a public IP address , type fw-pip for the name, and select OK

6. Select **Next**.

7. On the **IP addresses** tab, configure the following settings:

 Expand table

Setting	Value
Address space	Accept the default 10.0.0.0/16
Subnets	Select default , change Name to Workload-SN , and set Starting address to 10.0.2.0/24 . Select Save .

8. Select **Review + create**.

9. Select **Create**.

Note

Azure Firewall uses public IPs as needed based on available ports. After randomly selecting a public IP to connect outbound from, it will only use the next available public IP after no more connections can be made from the current public IP. In scenarios with high traffic volume and throughput, it's recommended to use a NAT Gateway to provide outbound connectivity. SNAT ports are dynamically allocated across all public IPs associated with NAT Gateway. To learn more, see [Scale SNAT ports with Azure NAT Gateway](#).

Create a virtual machine

Now create the workload virtual machine, and place it in the **Workload-SN** subnet.

- 1. On the Azure portal menu or from the **Home** page, select **Create a resource**.
- 2. Select **Ubuntu Server 22.04 LTS**.
- 3. Enter these values for the virtual machine:

 Expand table

Setting	Value
Resource group	Test-FW-RG
Virtual machine name	Srv-Work
Region	West US
Image	Ubuntu Server 22.04 LTS - x64 Gen2
Size	Standard_B2s
Authentication type	SSH public key
Username	azureuser
SSH public key source	Generate new key pair
Key pair name	Srv-Work_key

- 4. On the **Networking** tab, configure the following settings:

 Expand table

Setting	Value
Virtual network	Test-FW-VN
Subnet	Workload-SN
Public IP	None

- 5. Accept the other defaults and select **Next: Management**.
- 6. Accept the defaults and select **Next: Monitoring**.
- 7. For **Boot diagnostics**, select **Disable**. Accept the other defaults and select **Review + create**.

8. Review the settings on the summary page, and then select **Create**.
9. On the **Generate new key pair** dialog, select **Download private key and create resource**. Save the key file as **Srv-Work_key.pem**.
10. After the deployment is complete, select **Go to resource** and note the **Srv-Work** private IP address that you'll need to use later.

ⓘ Note

Azure provides a default outbound access IP for VMs that either aren't assigned a public IP address or are in the backend pool of an internal basic Azure load balancer. The default outbound access IP mechanism provides an outbound IP address that isn't configurable.

The default outbound access IP is disabled when one of the following events happens:

- A public IP address is assigned to the VM.
- The VM is placed in the backend pool of a standard load balancer, with or without outbound rules.
- An [Azure NAT Gateway](#) resource is assigned to the subnet of the VM.

VMs that you create by using virtual machine scale sets in flexible orchestration mode don't have default outbound access.

For more information about outbound connections in Azure, see [Default outbound access in Azure](#) and [Use Source Network Address Translation \(SNAT\) for outbound connections](#).

Examine the firewall

1. Go to the resource group and select the firewall.
2. Note the firewall private and public IP addresses. You use these addresses later.

Create a default route

When you create a route for outbound and inbound connectivity through the firewall, a default route to 0.0.0.0/0 with the virtual appliance private IP as a next hop is sufficient. This directs any outgoing and incoming connections through the firewall. As an example, if the firewall is fulfilling a TCP-handshake and responding to an incoming request, then the response is directed to the IP address who sent the traffic. This is by design.

As a result, there's no need to create another user defined route to include the AzureFirewallSubnet IP range. This might result in dropped connections. The original default route is sufficient.

For the **Workload-SN** subnet, configure the outbound default route to go through the firewall.

1. On the Azure portal, search for and select **Route tables**.
2. Select **Create**.
3. On the **Basics** tab, configure the following settings:

[Expand table](#)

Setting	Value
Subscription	Select your subscription
Resource group	Test-FW-RG
Region	West US
Name	Firewall-route
Propagate gateway routes	Select your preference

4. Select **Review + create**.
5. Select **Create**.

After deployment completes, select **Go to resource**.

1. On the **Firewall-route** page, select **Settings** > **Subnets** and then select **Associate**.
2. For **Virtual network**, select **Test-FW-VN**.
3. For **Subnet**, select **Workload-SN**. Make sure that you select only the **Workload-SN** subnet for this route, otherwise your firewall won't work correctly.
4. Select **OK**.
5. Select **Routes** and then select **Add**.
6. Configure the route with the following settings:

[Expand table](#)

Setting	Value
Route name	fw-dg
Destination type	IP Addresses
Destination IP addresses/CIDR ranges	0.0.0.0/0
Next hop type	Virtual appliance. Azure Firewall is actually a managed service, but virtual appliance works in this situation.
Next hop address	The private IP address for the firewall that you noted previously

7. Select **Add**.

Configure an application rule

This is the application rule that allows outbound access to `www.google.com`.

1. Open the **Test-FW-RG** resource group and select the **Test-FW01** firewall.
2. On the **Test-FW01** page, under **Settings**, select **Rules (classic)**.
3. Select the **Application rule collection** tab.
4. Select **Add application rule collection**.
5. Configure the rule collection with the following settings:

 Expand table

Setting	Value
Name	App-Coll01
Priority	200
Action	Allow

6. Under **Rules**, **Target FQDNs**, configure the following settings:

 Expand table

Setting	Value
Name	Allow-Google

Setting	Value
Source type	IP address
Source	10.0.2.0/24
Protocol:port	http, https
Target FQDNs	www.google.com

7. Select **Add**.

Azure Firewall includes a built-in rule collection for infrastructure FQDNs that are allowed by default. These FQDNs are specific for the platform and can't be used for other purposes. For more information, see [Infrastructure FQDNs](#).

Configure a network rule

This is the network rule that allows outbound access to two IP addresses at port 53 (DNS).

1. Select the **Network rule collection** tab.
2. Select **Add network rule collection**.
3. Configure the rule collection with the following settings:

 Expand table

Setting	Value
Name	Net-Coll01
Priority	200
Action	Allow

4. Under **Rules**, **IP addresses**, configure the following settings:

 Expand table

Setting	Value
Name	Allow-DNS
Protocol	UDP
Source type	IP address

Setting	Value
Source	10.0.2.0/24
Destination type	IP address
Destination address	209.244.0.3,209.244.0.4 (public DNS servers operated by Level3)
Destination Ports	53

5. Select **Add**.

Deploy Azure Bastion

Now deploy Azure Bastion to provide secure access to the virtual machine.

1. On the Azure portal menu, select **Create a resource**.
2. In the search box, type **Bastion** and select it from the results.
3. Select **Create**.
4. On the **Basics** tab, configure the following settings:

 Expand table

Setting	Value
Subscription	Select your subscription
Resource group	Test-FW-RG
Name	Test-Bastion
Region	West US
Tier	Developer
Virtual network	Test-FW-VN
Subnet	Select Edit subnet

5. In the **Edit subnet** page, configure the following settings:

 Expand table

Setting	Value
Starting address	10.0.4.0/26
Size	/26

6. Select **Save** and close the subnets page.

7. Select **Review + create**.

8. After validation passes, select **Create**.

Note

The Bastion deployment takes about 10 minutes to complete. The Developer tier is intended for test and evaluation purposes. For production deployments, review the Azure Bastion SKU options in [Azure Bastion SKU comparison](#).

Change the primary and secondary DNS address for the Srv-Work network interface

For testing purposes, configure the server's primary and secondary DNS addresses. This isn't a general Azure Firewall requirement.

1. On the Azure portal menu, select **Resource groups** or search for and select *Resource groups* from any page. Select the **Test-FW-RG** resource group.
2. Select the network interface for the **Srv-Work** virtual machine.
3. Under **Settings**, select **DNS servers**.
4. Under **DNS servers**, select **Custom**.
5. Configure the following DNS servers:

 Expand table

DNS server	Value
Primary	209.244.0.3
Secondary	209.244.0.4

6. Select **Save**.
7. Restart the **Srv-Work** virtual machine.

Test the firewall

Now, test the firewall to confirm that it works as expected.

1. In the Azure portal, navigate to the **Srv-Work** virtual machine.
2. Select **Connect**, then select **Connect via Bastion**.
3. Select **Use SSH Private Key from Local File**.
4. For **Username**, type **azureuser**.
5. Select the folder icon and browse to the **Srv-Work_key.pem** file you downloaded earlier.
6. Select **Connect**.
7. At the bash prompt, run the following commands to test DNS resolution:

Bash

```
nslookup www.google.com  
nslookup www.microsoft.com
```

Both commands should return answers, showing that your DNS queries are getting through the firewall.

8. Run the following commands to test the application rule:

Bash

```
curl https://www.google.com  
curl https://www.microsoft.com
```

The **www.google.com** request should succeed, and you should see the HTML response.

The **www.microsoft.com** request should fail, showing that the firewall is blocking the request.

So now you verified that the firewall rules are working:

- You can connect to the virtual machine using Bastion and SSH.
- You can browse to the one allowed FQDN, but not to any others.

- You can resolve DNS names using the configured external DNS server.

Clean up resources

You can keep your firewall resources to continue testing, or if no longer needed, delete the **Test-FW-RG** resource group to delete all firewall-related resources.

Next steps

- [Tutorial: Monitor Azure Firewall logs](#)
- [Learn more about Azure network security](#)

Last updated on 02/06/2026

What is Azure Firewall Manager?

Azure Firewall Manager is a security management service that provides central security policy and route management for cloud-based security perimeters.

Firewall Manager can provide security management for two network architecture types:

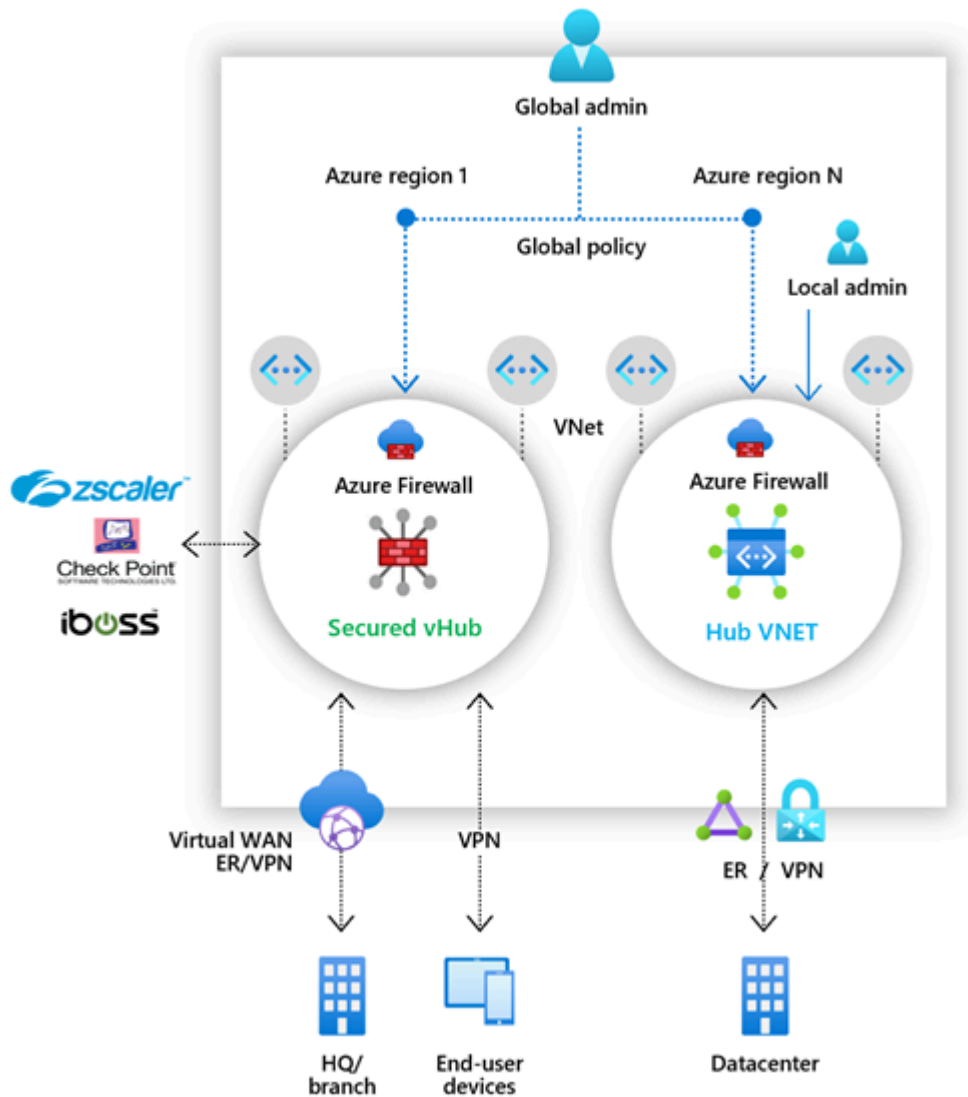
- **Secured virtual hub**

An [Azure Virtual WAN Hub](#) is a Microsoft-managed resource that lets you easily create hub and spoke architectures. When security and routing policies are associated with such a hub, it's referred to as a *secured virtual hub*.

- **Hub virtual network**

This is a standard Azure virtual network that you create and manage yourself. When security policies are associated with such a hub, it's referred to as a *hub virtual network*. At this time, only Azure Firewall Policy is supported. You can peer spoke virtual networks that contain your workload servers and services. You can also manage firewalls in standalone virtual networks that aren't peered to any spoke.

For a detailed comparison of *secured virtual hub* and *hub virtual network* architectures, see [What are the Azure Firewall Manager architecture options?](#)



Azure Firewall Manager features

Azure Firewall Manager offers the following features:

Central Azure Firewall deployment and configuration

You can centrally deploy and configure multiple Azure Firewall instances that span different Azure regions and subscriptions.

Hierarchical policies (global and local)

You can use Azure Firewall Manager to centrally manage Azure Firewall policies across multiple secured virtual hubs. Your central IT teams can author global firewall policies to enforce organization wide firewall policy across teams. Locally authored firewall policies allow a DevOps self-service model for better agility.

Integrated with partner security-as-a-service for advanced security

In addition to Azure Firewall, you can integrate partner security as a service (SECaaS) providers to provide more network protection for your virtual network and branch Internet connections.

This feature is available only with secured virtual hub deployments.

- Virtual network to Internet (V2I) traffic filtering
 - Filter outbound virtual network traffic with your preferred partner security provider.
 - Use advanced user-aware Internet protection for your cloud workloads running on Azure.
- Branch to Internet (B2I) traffic filtering

Use your Azure connectivity and global distribution to easily add partner filtering for branch to Internet scenarios.

For more information about security partner providers, see [What are Azure Firewall Manager security partner providers?](#)

Centralized route management

Easily route traffic to your secured hub for filtering and logging without the need to manually set up User Defined Routes (UDR) on spoke virtual networks.

This feature is available only with secured virtual hub deployments.

You can use partner providers for Branch to Internet (B2I) traffic filtering, side by side with Azure Firewall for Branch to virtual network (B2V), virtual network to virtual network (V2V) and virtual network to Internet (V2I).

DDoS protection plan

You can associate your virtual networks with a DDoS protection plan within Azure Firewall Manager. For more information, see [Configure an Azure DDoS Protection Plan using Azure Firewall Manager](#).

Manage Web Application Firewall policies

You can centrally create and associate Web Application Firewall (WAF) policies for your application delivery platforms, including Azure Front Door and Azure Application Gateway. For more information, see [Manage Web Application Firewall policies](#).

Region availability

Azure Firewall Policies can be used across regions. For example, you can create a policy in West US, and use it in East US.

Known issues

Azure Firewall Manager has the following known issues:

 Expand table

Issue	Description	Mitigation
Traffic splitting	Microsoft 365 and Azure Public PaaS traffic splitting isn't currently supported. As such, selecting a partner provider for V2I or B2I also sends all Azure Public PaaS and Microsoft 365 traffic via the partner service.	Investigating traffic splitting at the hub.
Base policies must be in same region as local policy	Create all your local policies in the same region as the base policy. You can still apply a policy that was created in one region on a secured hub from another region.	Investigating
Filtering inter-hub traffic in secure virtual hub deployments	Secured Virtual Hub to Secured Virtual Hub communication filtering is supported with the Routing Intent feature.	Enable Routing Intent on your Virtual WAN Hub by setting Inter-hub to Enabled in Azure Firewall Manager. See Routing Intent documentation for more information about this feature. The only Virtual WAN routing configuration that enables inter-hub traffic filtering is Routing intent.
Branch to branch traffic with private traffic filtering enabled	Branch to branch traffic can be inspected by Azure Firewall in secured hub scenarios if Routing Intent is enabled.	Enable Routing Intent on your Virtual WAN Hub by setting Inter-hub to Enabled in Azure Firewall Manager. See Routing Intent documentation for more information about this feature. The only Virtual WAN routing configuration that enables branch to branch private traffic is routing intent.

Issue	Description	Mitigation
All Secured Virtual Hubs sharing the same virtual WAN must be in the same resource group.	This behavior is aligned with Virtual WAN Hubs today.	Create multiple Virtual WANs to allow Secured Virtual Hubs to be created in different resource groups.
Bulk IP address addition fails	The secure hub firewall goes into a failed state if you add multiple public IP addresses.	Add smaller public IP address increments. For example, add 10 at a time.
DDoS Protection not supported with secured virtual hubs	DDoS Protection isn't integrated with vWANs.	Investigating You can now associate a public IP address created in your tenants to secured hubs. The public IP addresses can be configured with Azure DDoS Protection. For more information, see Customer provided public IP address support in secured hubs (preview) .
Activity logs not fully supported	Firewall policy doesn't currently support Activity logs.	Investigating
Description of rules not fully supported	Firewall policy doesn't display the description of rules in an ARM export.	Investigating
Azure Firewall Manager overwrites static and custom routes causing downtime in virtual WAN hub.	You shouldn't use Azure Firewall Manager to manage your settings in deployments configured with custom or static routes. Updates from Firewall Manager can potentially overwrite static or custom route settings.	If you use static or custom routes, use the Virtual WAN page to manage security settings and avoid configuration via Azure Firewall Manager. For more information, see Scenario: Azure Firewall - custom .

Next steps

- [Learn module: Introduction to Azure Firewall Manager](#)
- Review [Azure Firewall Manager deployment overview](#)
- Learn about [secured Virtual Hubs](#)
- [Learn more about Azure network security](#)

Last updated on 01/27/2025

What is Azure Firewall?

Azure Firewall is a cloud-native, intelligent network firewall security service that offers top-tier threat protection for your Azure cloud workloads. It's a fully stateful firewall as a service, featuring built-in high availability and unlimited cloud scalability. Azure Firewall inspects both east-west and north-south traffic. To understand these traffic types, see [east-west and north-south traffic](#).

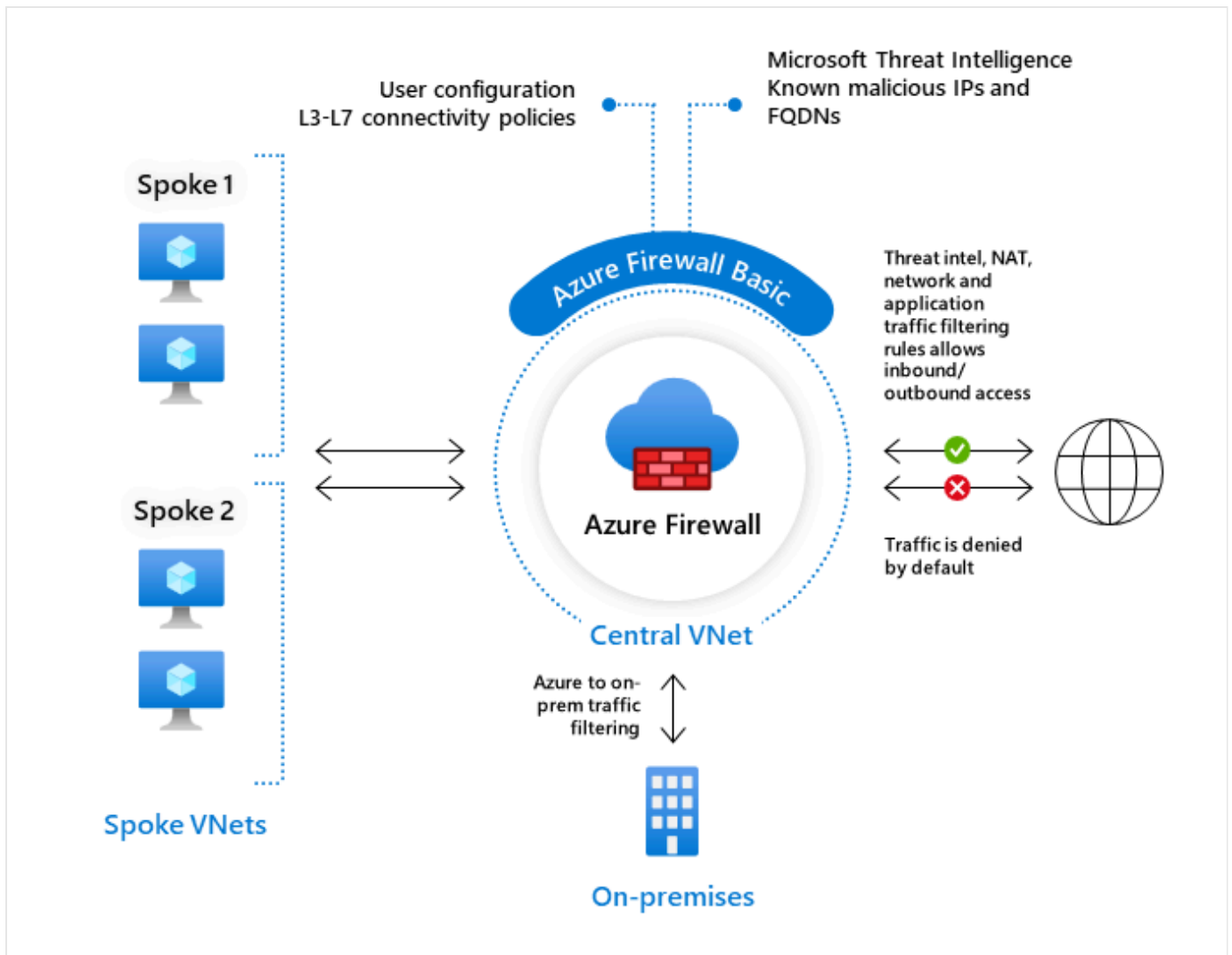
Azure Firewall is available in three SKUs: Basic, Standard, and Premium.

ⓘ Note

Azure Firewall is one of the services that make up the Network Security category in Azure. Other services in this category include [Azure DDoS Protection](#) and [Azure Web Application Firewall](#). Each service has its own unique features and use cases. For more information on this service category, see [Network Security](#).

Azure Firewall Basic

Azure Firewall Basic is designed for small and medium-sized businesses (SMBs) to secure their Azure cloud environments. It provides essential protection at an affordable price.



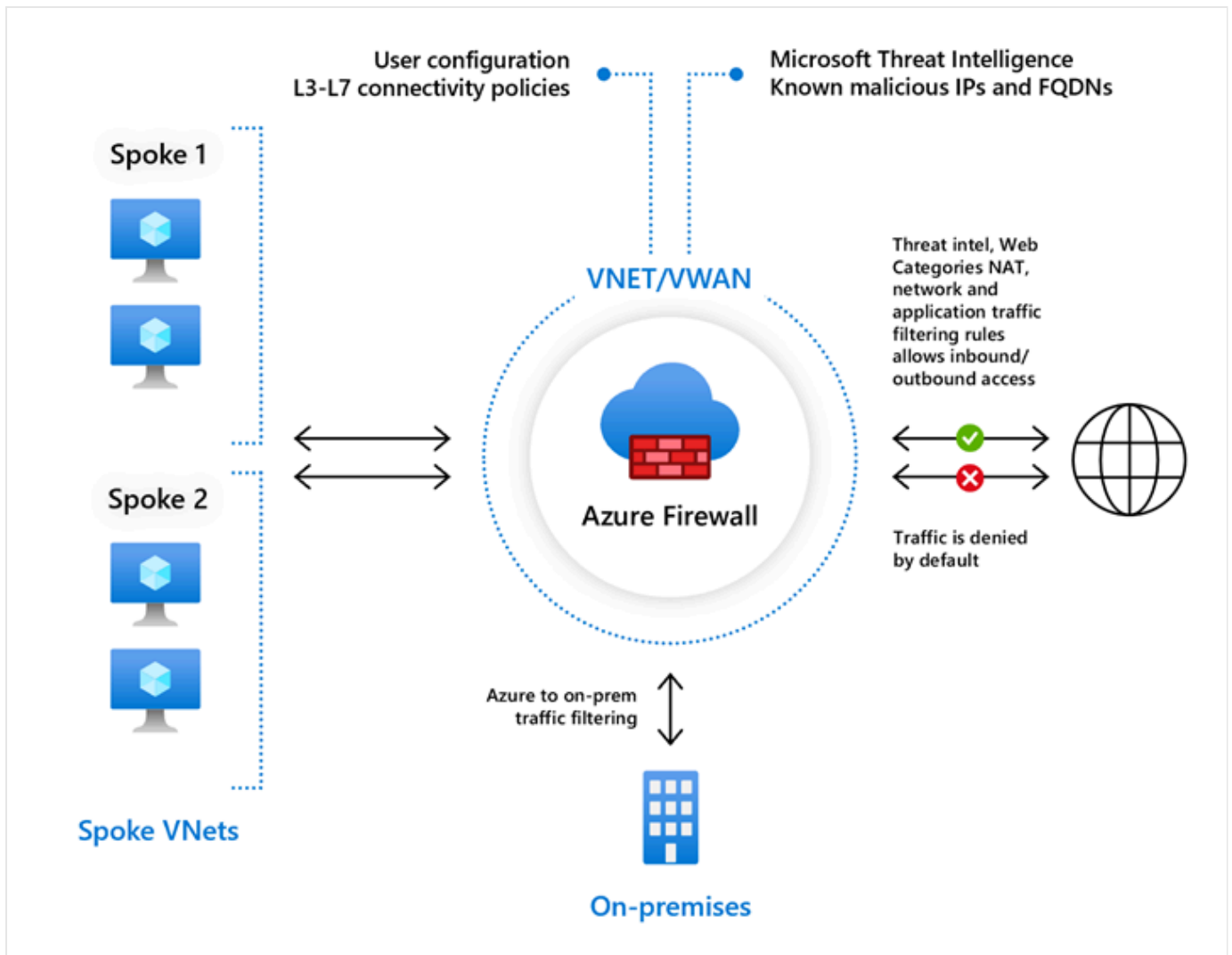
Key limitations of Azure Firewall Basic include:

- Supports Threat Intel *alert mode* only
- Fixed scale unit with two virtual machine backend instances
- Recommended for environments with an estimated throughput of 250 Mbps

For more information, see [Azure Firewall features by SKU](#).

Azure Firewall Standard

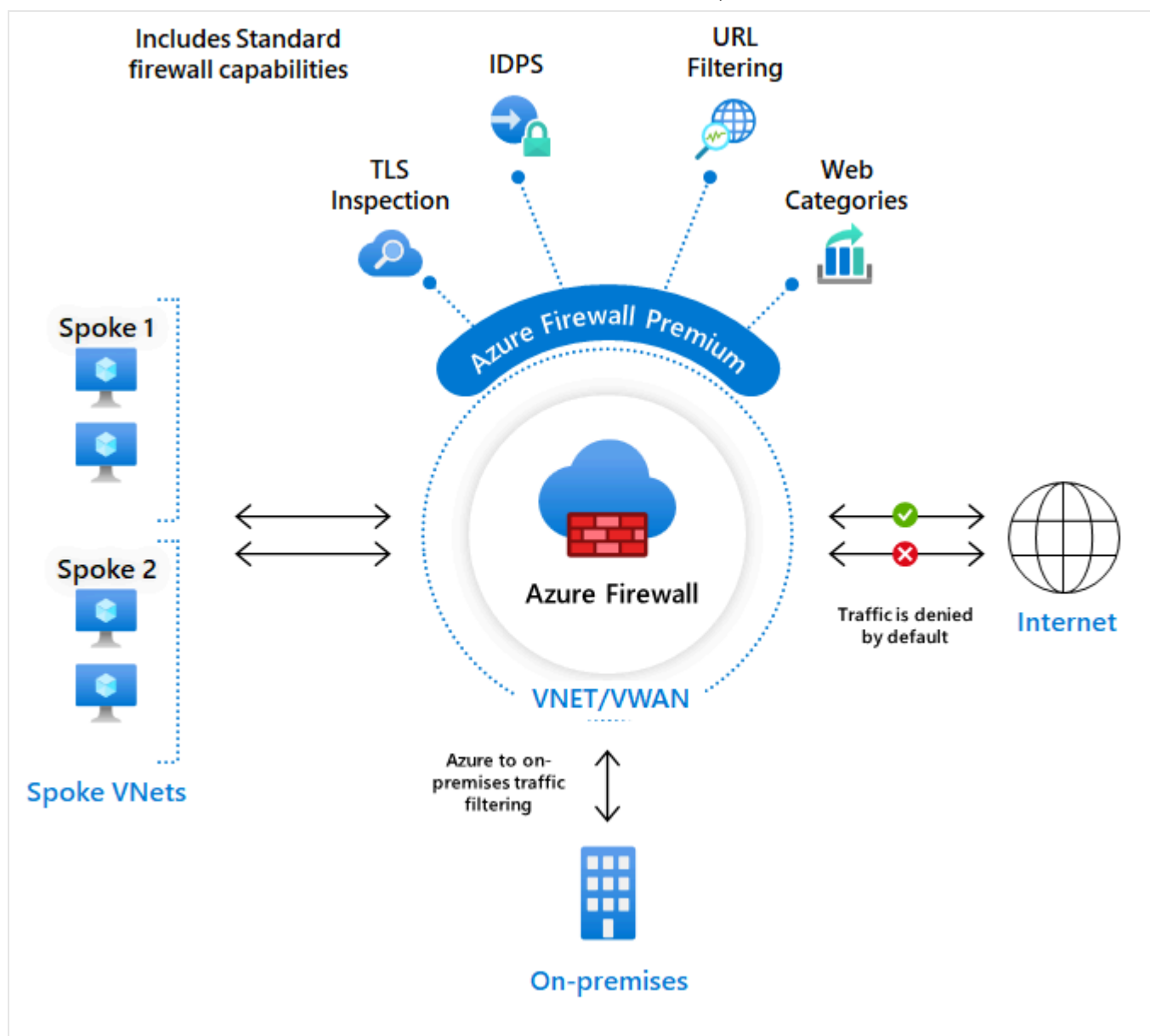
Azure Firewall Standard offers L3-L7 filtering and threat intelligence feeds directly from Microsoft Cyber Security. It can alert you to and block traffic from or to known malicious IP addresses and domains. It updates in real time to protect against new and emerging threats.



For more information, see [Azure Firewall features by SKU](#).

Azure Firewall Premium

Azure Firewall Premium provides advanced capabilities, including signature-based IDPS for rapid attack detection by identifying specific patterns. These patterns can include byte sequences in network traffic or known malicious instruction sequences used by malware. With over 67,000 signatures in more than 50 categories, updated in real time, it protects against new and emerging exploits such as malware, phishing, coin mining, and Trojan attacks.



For more information, see [Azure Firewall features by SKU](#).

Feature comparison

To compare all Azure Firewall SKU features, see [Choose the right Azure Firewall SKU to meet your needs](#).

Azure Firewall Manager

Azure Firewall Manager provides a central place to manage Azure Firewalls across multiple subscriptions. It uses firewall policies to apply a common set of network and application rules and configurations to the firewalls in your tenant.

Firewall Manager supports firewalls in both virtual network and Virtual WAN (Secure Virtual Hub) environments. Secure Virtual Hubs use the Virtual WAN route automation solution to simplify

routing traffic to the firewall with just a few steps.

To learn more, see [Azure Firewall Manager](#).

Pricing and SLA

For pricing details, see [Azure Firewall pricing](#) .

For SLA information, see [Service Level Agreements \(SLA\) for Online Services](#) .

Supported regions

For a list of supported regions, see [Azure products available by region](#) .

What's new

To learn about the latest updates, see [Azure updates](#) .

Known problems

For known problems, see [Azure Firewall known issues](#).

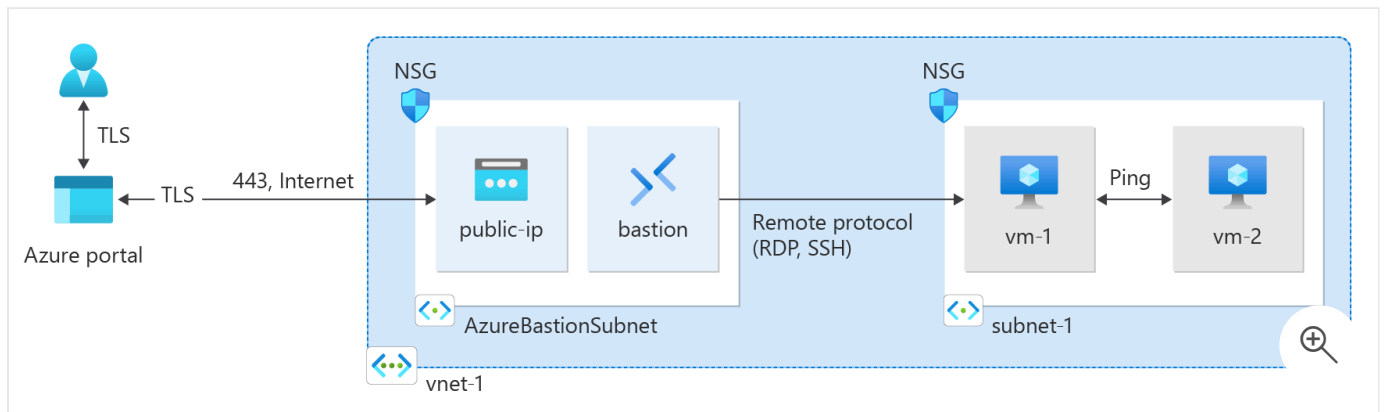
Next steps

- [Quickstart: Create an Azure Firewall and a firewall policy - ARM template](#)
- [Quickstart: Deploy Azure Firewall with Availability Zones - ARM template](#)
- [Tutorial: Deploy and configure Azure Firewall by using the Azure portal](#)
- [Learn module: Introduction to Azure Firewall](#)
- [Learn more about Azure network security](#)

Last updated on 03/29/2026

Quickstart: Create an Azure Virtual Network

Learn how to create an Azure Virtual Network using the Azure portal, Azure CLI, Azure PowerShell, Azure Resource Manager (ARM) template, Bicep template, and Terraform. Two virtual machines and an Azure Bastion host are deployed to test connectivity between the virtual machines in the same virtual network. The Azure Bastion host facilitates secure and seamless RDP and SSH connectivity to the virtual machines directly in the Azure portal over SSL.



A virtual network is the fundamental building block for private networks in Azure. Azure Virtual Network enables Azure resources such as virtual machines to securely communicate with each other and the internet.

<https://learn-video.azurefd.net/vod/player?id=6b5b138e-8406-406e-8b34-40bdadf9fc6d&locale=en-us&embedUrl=%2Fazure%2Fvirtual-network%2Fquickstart-create-virtual-network>

If you don't have an Azure account with an active subscription, [create one for free](#).

Prerequisites

Portal

- An Azure account with an active subscription. You can [create an account for free](#).

Resource values

Use the following values to replace the placeholders of resources in this article:

 Expand table

Setting	Placeholder	Value
Resource group	<resource-group>	test-rg
Region	<region>	East US 2
Virtual network	<virtual-network>	vnet-1
Subnet	<subnet>	subnet-1
Network security group	<network-security-group>	nsg-1
Bastion	<bastion>	bastion
Virtual machine 1	<virtual-machine-1>	vm-1
Virtual machine 2	<virtual-machine-2>	vm-2

Portal

Create a resource group

1. Sign in to the [Azure portal](#) with your Azure account.
2. In the search box at the top of the portal, enter **Resource group**. Select **Resource groups** in the search results.
3. Select + **Create**.
4. In the **Basics** tab of **Create a resource group**, enter, or select the following information:

 Expand table

Setting	Value
Subscription	Select your subscription.
Resource group	Enter <resource-group>.
Region	Select <region>.

5. Select **Review + create**.
6. Select **Create**.

Create a virtual network

1. In the search box at the top of the portal, enter **Virtual network**. Select **Virtual networks** in the search results.
2. Select **+ Create**.
3. On the **Basics** tab of **Create virtual network**, enter, or select the following information:

[Expand table](#)

Setting	Value
Project details	
Subscription	Select your subscription.
Resource group	Select <resource-group>.
Instance details	
Name	Enter <virtual-network>.
Region	Select <region>.

4. Select **Next** to proceed to the **Security** tab.
5. Select **Next** to proceed to the **IP Addresses** tab.
6. In the address space box in **Subnets**, select the **default** subnet.
7. In **Edit subnet**, enter, or select the following information:

[Expand table](#)

Setting	Value
Subnet details	
Subnet template	Leave the default Default .
Name	Enter <subnet>.
Starting address	Leave the default of 10.0.0.0 .
Subnet size	Leave the default of /24 (256 addresses) .

8. Select **Save**.

9. Select **Review + create** at the bottom of the screen, and when validation passes, select **Create**.

Deploy Azure Bastion

Azure Bastion uses your browser to connect to virtual machines (VMs) in your virtual network over secure shell (SSH) or remote desktop protocol (RDP) by using their private IP addresses. The virtual machines don't need public IP addresses, client software, or special configuration. For more information about Azure Bastion, see [Azure Bastion](#).

ⓘ **Note**

Hourly pricing starts from the moment that Bastion is deployed, regardless of outbound data usage. For more information, see [Pricing](#) and [SKUs](#). If you're deploying Bastion as part of a tutorial or test, we recommend that you delete this resource after you finish using it.

- 1. In the search box at the top of the portal, enter **Bastion**. Select **Bastions** in the search results.
- 2. Select **+ Create**.
- 3. In the **Basics** tab of **Create a Bastion**, enter, or select the following information:

 **Expand table**

Setting	Value
Project details	
Subscription	Select your subscription.
Resource group	Select <resource-group>.
Instance details	
Name	Enter <bastion>.
Region	Select <region>.
Tier	Select Developer .
Configure virtual networks	

Setting	Value
Virtual network	Select <virtual-network>.

- 4. Select **Review + create**.
- 5. Select **Create**.

Create virtual machines

The following procedure creates two VMs named <virtual-machine-1> and <virtual-machine-2> in the virtual network:

- 1. In the portal, search for and select **Virtual machines**.
- 2. In **Virtual machines**, select + **Create**, and then select **Azure virtual machine**.
- 3. On the **Basics** tab of **Create a virtual machine**, enter, or select the following information:

 Expand table

Setting	Value
Project details	
Subscription	Select your subscription.
Resource group	Select <resource-group>.
Instance details	
Virtual machine name	Enter <virtual-machine-1>.
Region	Select <region>.
Availability options	Select No infrastructure redundancy required .
Security type	Leave the default of Standard .
Image	Select Ubuntu Server 22.04 LTS - x64 Gen2 .
VM architecture	Leave the default of x64 .
Size	Select a size.
Administrator account	

Setting	Value
Authentication type	Select SSH public key .
Username	Enter azureuser .
SSH public key source	Select Generate new key pair .
Key pair name	Enter <virtual-machine-1>-key .
Inbound port rules	
Public inbound ports	Select None .

4. Select the **Networking** tab. Enter or select the following information:

 Expand table

Setting	Value
Network interface	
Virtual network	Select <virtual-network> .
Subnet	Select <subnet> (10.0.0.0/24) .
Public IP	Select None .
NIC network security group	Select Advanced .
Configure network security group	Select Create new . Enter <network-security-group> for the name. Leave the rest at the defaults and select OK .

5. Leave the rest of the settings at the defaults and select **Review + create**.

6. Review the settings and select **Create**.

7. Wait for the first virtual machine to deploy then repeat the previous steps to create a second virtual machine with the following settings:

 Expand table

Setting	Value
Virtual machine name	Enter <virtual-machine-2> .
SSH public key source	Select Generate new key pair .

Setting	Value
Key pair name	Enter <virtual-machine-2>-key.
Virtual network	Select <virtual-network>.
Subnet	Select <subnet> (10.0.0.0/24).
Public IP	Select None .
NIC network security group	Select Advanced .
Configure network security group	Select <network-security-group>.

ⓘ Note

Virtual machines in a virtual network with an Azure Bastion host don't need public IP addresses. Bastion provides the public IP, and the VMs use private IPs to communicate within the network. You can remove the public IPs from any VMs in Bastion-hosted virtual networks. For more information, see [Dissociate a public IP address from an Azure VM](#).

ⓘ Note

Azure provides a default outbound access IP for VMs that either aren't assigned a public IP address or are in the backend pool of an internal basic Azure load balancer. The default outbound access IP mechanism provides an outbound IP address that isn't configurable.

The default outbound access IP is disabled when one of the following events happens:

- A public IP address is assigned to the VM.
- The VM is placed in the backend pool of a standard load balancer, with or without outbound rules.
- An [Azure NAT Gateway](#) resource is assigned to the subnet of the VM.

VMs that you create by using virtual machine scale sets in flexible orchestration mode don't have default outbound access.

For more information about outbound connections in Azure, see [Default outbound access in Azure](#) and [Use Source Network Address Translation \(SNAT\) for outbound connections](#).

Connect to a virtual machine

1. In the search box at the top of the portal, enter **Virtual machine**. Select **Virtual machines** in the search results.
2. In **Virtual machines**, select **vm-1**.
3. Select **Connect** then **Connect via Bastion** in the **Overview** section.
4. In the **Bastion** connection page, enter or select the following information:

 Expand table

Setting	Value
Authentication Type	Select SSH Private Key from Local File.
Username	Enter azureuser .
Local File	Select the private key file you downloaded or created.

5. Select **Connect**.

Start communication between virtual machines

1. At the bash prompt for **vm-1**, enter `ping -c 4 vm-2`.

You get a reply similar to the following message:

Output

```
azureuser@vm-1:~$ ping -c 4 vm-2
PING vm-2.3bnkevn3313ujpr5l1kqop4n4d.cx.internal.cloudapp.net (10.0.0.5) 56(84)
bytes of data.
64 bytes from vm-2.internal.cloudapp.net (10.0.0.5): icmp_seq=1 ttl=64 time=1.83
ms
64 bytes from vm-2.internal.cloudapp.net (10.0.0.5): icmp_seq=2 ttl=64
time=0.987 ms
64 bytes from vm-2.internal.cloudapp.net (10.0.0.5): icmp_seq=3 ttl=64
time=0.864 ms
```

```
64 bytes from vm-2.internal.cloudapp.net (10.0.0.5): icmp_seq=4 ttl=64
time=0.890 ms
```

2. Close the Bastion session.
3. In the search box at the top of the portal, enter **Virtual machine**. Select **Virtual machines** in the search results.
4. In **Virtual machines**, select **vm-2**.
5. Select **Connect** then **Connect via Bastion** in the **Overview** section.
6. In the **Bastion** connection page, enter or select the following information:

[Expand table](#)

Setting	Value
Authentication Type	Select SSH Private Key from Local File.
Username	Enter azureuser.
Local File	Select the private key file you downloaded or created.

7. Select **Connect**.
8. At the bash prompt for **vm-2**, enter `ping -c 4 vm-1`.

You get a reply similar to the following message:

Output

```
azureuser@vm-2:~$ ping -c 4 vm-1
PING vm-1.3bnkevn3313ujpr5l1kqop4n4d.cx.internal.cloudapp.net (10.0.0.4) 56(84)
bytes of data.
64 bytes from vm-1.internal.cloudapp.net (10.0.0.4): icmp_seq=1 ttl=64
time=0.695 ms
64 bytes from vm-1.internal.cloudapp.net (10.0.0.4): icmp_seq=2 ttl=64
time=0.896 ms
64 bytes from vm-1.internal.cloudapp.net (10.0.0.4): icmp_seq=3 ttl=64 time=3.43
ms
64 bytes from vm-1.internal.cloudapp.net (10.0.0.4): icmp_seq=4 ttl=64
time=0.780 ms
```

9. Close the Bastion session.

Clean up resources

Portal

When you finish using the resources that you created, you can delete the resource group and all its resources.

1. In the Azure portal, search for and select **Resource groups**.
2. On the **Resource groups** page, select the **<resource-group>** resource group.
3. On the **<resource-group>** page, select **Delete resource group**.
4. Enter **<resource-group>** in **Enter resource group name to confirm deletion**, and then select **Delete**.

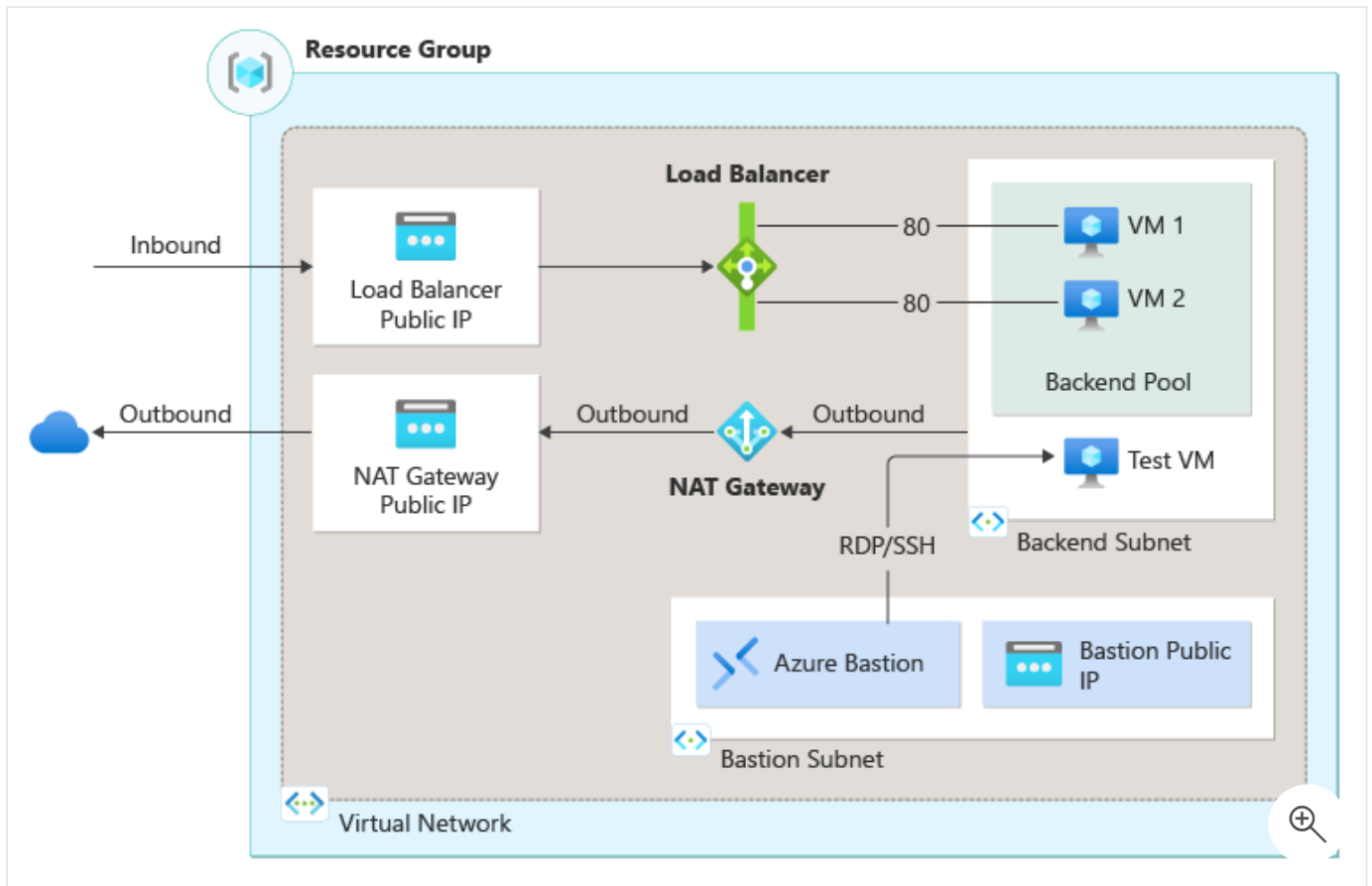
Related content

- [Filter network traffic](#)
- [Learn more about using Terraform on Azure](#)

Last updated on 03/27/2026

Quickstart: Create a public load balancer to load balance VMs using the Azure portal

Get started with Azure Load Balancer by using the Azure portal to create a public load balancer for a backend pool with two virtual machines. Other resources include Azure Bastion, NAT Gateway, a virtual network, and the required subnets.



Prerequisites

- An Azure account with an active subscription. [Create an account for free](#) .

Sign in to Azure

Sign in to the [Azure portal](#) .

Create NAT gateway

In this section, you create a NAT gateway for outbound internet access for resources in the virtual network. For other options for outbound rules, check out [Network Address Translation \(SNAT\) for](#)

outbound connections

1. In the search box at the top of the portal, enter **NAT gateway**. Select **NAT gateways** in the search results.
2. Select **+ Create**.
3. In the **Basics** tab of **Create network address translation (NAT) gateway** enter or select the following information:

 Expand table

Setting	Value
Project details	
Subscription	Select your subscription.
Resource group	Select Create a new resource group . Enter load-balancer-rg in Name. Select OK .
Instance details	
NAT gateway name	Enter lb-nat-gateway .
Region	Select East US .
SKU	Select Standard .
Availability zone	Select None .
TCP idle timeout (minutes)	Enter 15 .

4. Select the **Outbound IP** tab or select the **Next** button at the bottom of the page.
5. Select **Add public IP addresses or prefixes**.
6. On the **Manage public IP addresses and prefixes** page, select **Create a public IP address** and enter **nat-gw-public-ip** in Name.
7. Select **OK** and **Save**.
8. Select the blue **Review + create** button at the bottom of the page, or select the **Review + create** tab.
9. Select **Create**.

Create a virtual network and bastion host

In this section, you create a virtual network with a resource subnet, an Azure Bastion subnet, and an Azure Bastion host.

 Important

Hourly pricing starts from the moment that Bastion is deployed, regardless of outbound data usage. For more information, see [Pricing](#) and [SKUs](#). If you're deploying Bastion as part of a tutorial or test, we recommend that you delete this resource after you finish using it.

- 1. In the portal, search for and select **Virtual networks**.
- 2. On the **Virtual networks** page, select **+ Create**.
- 3. On the **Basics** tab of **Create virtual network**, enter or select the following information:

 **Expand table**

Setting	Value
Project details	
Subscription	Select your subscription.
Resource group	Select load-balancer-rg from the dropdown or Create new if it doesn't exist. Enter load-balancer-rg in Name. Select OK .
Instance details	
Name	Enter lb-vnet .
Region	Select (US) East US .

- 4. Select the **Security** tab or **Next** button at the bottom of the page.
- 5. Under **Azure Bastion**, enter or select the following information:

 **Expand table**

Setting	Value
Azure Bastion	
Enable Azure Bastion	Select checkbox.
Azure Bastion host name	Enter lb-vnet-bastion .
Azure Bastion public IP address	Select Create new . Enter lb-vnet-bastion-ip in Name. Select OK .

6. Select the **IP addresses** tab, or **Next** at the bottom of the page.

7. On **Create virtual network** page, enter or select the following information:

 Expand table

Setting	Value
IPv4 address space	Enter 10.0.0.0/16 (65,356 addresses) .
Subnets	Select the default subnet link to edit.
Edit subnet	
Subnet purpose	Leave the default Default .
Name	Enter backend-subnet .
Starting address	Enter 10.0.0.0 .
Subnet size	Enter /24(256 addresses) .
Security	
NAT Gateway	Select lb-nat-gateway .

8. Select **Save**.

9. Select **Review + create** at the bottom of the screen, and when validation passes, select **Create**.

Create load balancer

In this section, you create a zone redundant load balancer that load balances virtual machines. With zone-redundancy, one or more availability zones can fail and the data path survives as long as one zone in the region remains healthy.

During the creation of the load balancer, you configure:

- Frontend IP address
- Backend pool
- Inbound load-balancing rules
- Health probe

1. In the search box at the top of the portal, enter **Load balancer**. Select **Load balancers** in the search results.
2. In the **Load balancer** page, select + **Create**.
3. In the **Basics** tab of the **Create load balancer** page, enter or select the following information:

 Expand table

Setting	Value
Project details	
Subscription	Select your subscription
Resource group	Select load-balancer-rg
Instance details	
Name	Enter load-balancer
Region	Select East US
SKU	Leave the default Standard
Type	Select Public
Tier	Leave the default Regional

Create load balancer

Basics Frontend IP configuration Backend pools Inbound rules Outbound rules Tags Review + create

Azure load balancer is a layer 4 load balancer that distributes incoming traffic among healthy virtual machine instances. Load balancers uses a hash-based distribution algorithm. By default, it uses a 5-tuple (source IP, source port, destination IP, destination port, protocol type) hash to map traffic to available servers. Load balancers can either be internet-facing where it is accessible via public IP addresses, or internal where it is only accessible from a virtual network. Azure load balancers also support Network Address Translation (NAT) to route traffic between public and private IP addresses. [Learn more.](#)

Project details

Subscription * Contoso Subscription

Resource group * load-balancer-rg [Create new](#)

Instance details

Name * load-balancer ✓

Region * East US

SKU * ☒ Standard ☐ Gateway ☐ Basic (Retiring on September 30, 2025)

Type * ☒ Public ☐ Internal

Tier * ☒ Regional ☐ Global

Review + create

< Previous

Next : Frontend IP configuration >

[Download a template for automation](#) [Give feedback](#)

4. Select **Next: Frontend IP configuration** at the bottom of the page.
5. In **Frontend IP configuration**, select **+ Add a frontend IP configuration**.
6. Enter **lb-frontend** in **Name**.
7. Select **IPv4** for the **IP version**.
8. Select **IP address** for the **IP type**.

! Note

For more information on IP prefixes, see [Azure Public IP address prefix](#).

9. Select **Create new** in **Public IP address**.
10. In **Add a public IP address**, enter **lb-frontend-ip** for **Name**.
11. Select **Zone-redundant** in **Availability zone**.

Note

In regions with [Availability Zones](#), you can select zone-redundant (default option) or a specific zone. The choice depends on your specific domain failure requirements. In regions without Availability Zones, this field won't appear.

For more information on availability zones, see [Availability zones overview](#).

12. Leave the default of **Microsoft Network** for **Routing preference**.
13. Select **Save**.
14. Select **Save**.
15. Select **Next: Backend pools** at the bottom of the page.
16. In the **Backend pools** tab, select + **Add a backend pool**.
17. Enter **lb-backend-pool** for **Name** in **Add backend pool**.
18. Select **lb-vnet** in **Virtual network**.
19. Select **IP Address** for **Backend Pool Configuration**.
20. Select **Save**.
21. Select **Next: Inbound rules** at the bottom of the page.
22. Under **Load balancing rule** in the **Inbound rules** tab, select + **Add a load balancing rule**.
23. In **Add load balancing rule**, enter or select the following information:

[Expand table](#)

Setting	Value
Name	Enter lb-HTTP-rule
IP Version	Select IPv4 or IPv6 depending on your requirements
Frontend IP address	Select lb-frontend (To be created)
Backend pool	Select lb-backend-pool
Protocol	Select TCP
Port	Enter 80

Setting	Value
Backend port	Enter 80
Health probe	Select Create new . In Name , enter lb-health-probe . Select HTTP in Protocol . Leave the rest of the defaults, and select Save .
Session persistence	Select None .
Idle timeout (minutes)	Enter or select 15
Enable TCP reset	Select checkbox
Enable Floating IP	Leave unchecked
Outbound source network address translation (SNAT)	Leave the default of (Recommended) Use outbound rules to provide backend pool members access to the internet .

24. Select **Save**.

25. Select the blue **Review + create** button at the bottom of the page.

26. Select **Create**.

ⓘ Note

In this example we'll create a NAT gateway to provide outbound Internet access. The outbound rules tab in the configuration is bypassed as it's optional and isn't needed with the NAT gateway. For more information on Azure NAT gateway, see [What is Azure Virtual Network NAT?](#) For more information about outbound connections in Azure, see [Source Network Address Translation \(SNAT\) for outbound connections](#)

Create virtual machines

In this section, you create two VMs (**lb-vm1** and **lb-VM2**) in two different zones (**Zone 1** and **Zone 2**).

These VMs are added to the backend pool of the load balancer that was created earlier.

1. In the search box at the top of the portal, enter **Virtual machine**. Select **Virtual machines** in the search results.
2. In **Compute infrastructure | Virtual machines**, select **+ Create > Virtual machine**.

3. In **Create a virtual machine**, enter or select the following values in the **Basics** tab:

[Expand table](#)

Setting	Value
Project Details	
Subscription	Select your Azure subscription
Resource Group	Select load-balancer-rg
Instance details	
Virtual machine name	Enter lb-VM1
Region	Select ((US) East US)
Availability Options	Select Availability zones
Availability zone	Select Zone 1
Security type	Select Standard .
Image	Select Windows Server 2022 Datacenter: Azure Edition - x64 Gen2
Azure Spot instance	Leave the default of unchecked.
Size	Choose VM size or take default setting
Administrator account	
Username	Enter a username
Password	Enter a password
Confirm password	Reenter password
Inbound port rules	
Public inbound ports	Select None

4. Select the **Networking** tab, or select **Next: Disks**, then **Next: Networking**.

5. In the **Networking** tab, select or enter the following information:

[Expand table](#)

Setting	Value
Network interface	

Setting	Value
Virtual network	Select lb-vnet
Subnet	Select backend-subnet
Public IP	Select None .
NIC network security group	Select Advanced
Configure network security group	Skip this setting until the rest of the settings are completed. Complete after Select a backend pool .
Delete NIC when VM is deleted	Leave the default of unselected .
Accelerated networking	Leave the default of selected .
Load balancing	
Load balancing options	
Load-balancing options	Select Azure load balancer
Select a load balancer	Select load-balancer
Select a backend pool	Select lb-backend-pool
Configure network security group	Select Create new . In the Create network security group , enter lb-NSG in Name . Under Inbound rules , select +Add an inbound rule . In Service , select HTTP . Under Priority , enter 100 . In Name , enter lb-NSG-Rule Select Add Select Add then select OK .

6. Select **Review + create**.

7. Review the settings, and then select **Create**.

8. Follow the steps 1 through 7 to create another VM with the following values and all the other settings the same as **lb-VM1**:

[Expand table](#)

Setting	VM 2
Name	lb-VM2

Setting	VM 2
Availability zone	Zone 2
Network security group	Select the existing Ib-NSG

ⓘ Note

Azure provides a default outbound access IP for VMs that either aren't assigned a public IP address or are in the backend pool of an internal basic Azure load balancer. The default outbound access IP mechanism provides an outbound IP address that isn't configurable.

The default outbound access IP is disabled when one of the following events happens:

- A public IP address is assigned to the VM.
- The VM is placed in the backend pool of a standard load balancer, with or without outbound rules.
- An [Azure NAT Gateway](#) resource is assigned to the subnet of the VM.

VMs that you create by using virtual machine scale sets in flexible orchestration mode don't have default outbound access.

For more information about outbound connections in Azure, see [Default outbound access in Azure](#) and [Use Source Network Address Translation \(SNAT\) for outbound connections](#).

Install IIS

1. In the search box at the top of the portal, enter **Virtual machine**. Select **Virtual machines** in the search results.
2. Select **Ib-VM1**.
3. On the **Overview** page, select **Connect**, then **Bastion**.
4. Enter the username and password entered during VM creation.
5. Select **Connect**.
6. On the server desktop, navigate to **Start > Windows PowerShell > Windows PowerShell**.
7. In the PowerShell Window, run the following commands to:

- Install the IIS server.
- Remove the default iisstart.htm file.
- Add a new iisstart.htm file that displays the name of the VM:

PowerShell

```
# Install IIS server role
Install-WindowsFeature -name Web-Server -IncludeManagementTools

# Remove default htm file
Remove-Item C:\inetpub\wwwroot\iisstart.htm

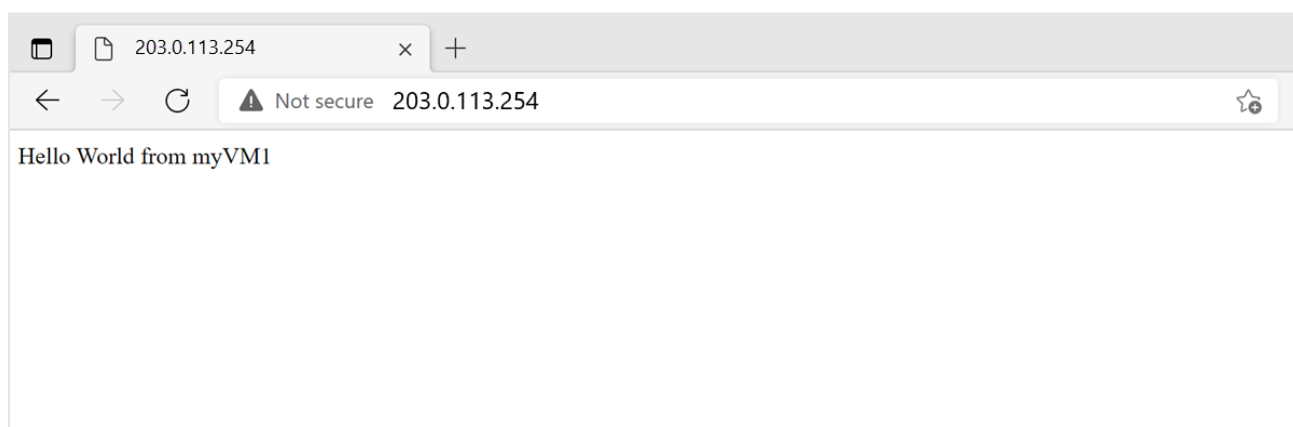
# Add a new htm file that displays server name
Add-Content -Path "C:\inetpub\wwwroot\iisstart.htm" -Value $("Hello World from
" + $env:computername)
```

8. Close the Bastion session with **lb-VM1**.

9. Repeat steps 1 to 8 to install IIS and the updated iisstart.htm file on **lb-VM2**.

Test the load balancer

1. In the search box at the top of the page, enter **Public IP**. Select **Public IP addresses** in the search results.
2. In **Public IP addresses**, select **frontend-ip**.
3. Copy the item in **IP address**. Paste the public IP into the address bar of your browser. The custom VM page of the IIS Web server is displayed in the browser.



Clean up resources

When no longer needed, delete the resource group, load balancer, and all related resources. To do so, select the resource group **load-balancer-rg** that contains the resources and then select **Delete**.

Next steps

In this quickstart, you:

- Created an Azure Load Balancer
- Attached 2 VMs to the load balancer
- Tested the load balancer

To learn more about Azure Load Balancer, continue to:

[What is Azure Load Balancer?](#)

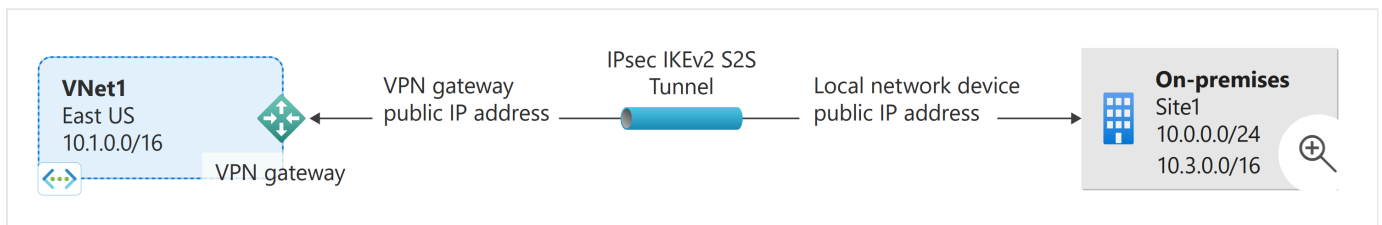
Last updated on 01/09/2026

VPN Gateway topology and design

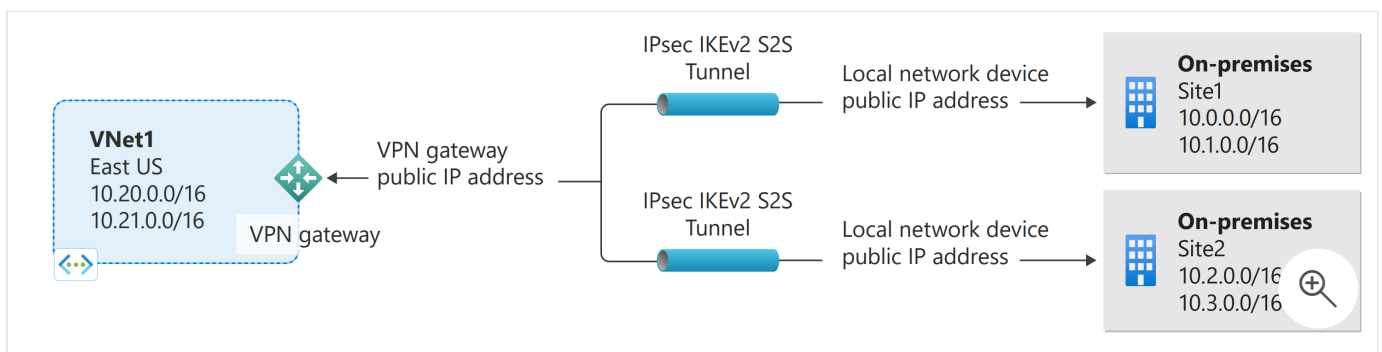
There are many different options available for virtual network connections. To help you select a VPN gateway connection topology that meets your requirements, use the diagrams and descriptions in the following sections. The diagrams show the main baseline topologies, but it's possible to build more complex configurations using the diagrams as guidelines.

Site-to-site VPN

A site-to-site (S2S) VPN gateway connection is a connection over IPsec/IKE (IKEv1 or IKEv2) VPN tunnel. Site-to-site connections can be used for cross-premises and hybrid configurations. A site-to-site connection requires a VPN device located on-premises that has a public IP address assigned to it.



You can create more than one VPN connection from your virtual network gateway, typically connecting to multiple on-premises sites. When working with multiple connections, you must use a RouteBased VPN type. Because each virtual network can only have one VPN gateway, all connections through the gateway share the available bandwidth. This type of connectivity design is sometimes referred to as *multi-site*.



If you want to create a design for highly available gateway connectivity, you can configure your gateway to be in active-active mode. This mode lets you configure two active tunnels (one from each gateway virtual machine instance) to the same VPN device to create highly available connectivity. In addition to being a highly available connectivity design, another advantage of active-active mode is that customers experience higher throughputs.

- For information about selecting a VPN device, see the [VPN Gateway FAQ - VPN devices](#).
- For information about highly available connections, see [Designing highly available connections](#).
- For information about active-active mode, see [About active-active mode gateways](#).

Deployment methods for S2S

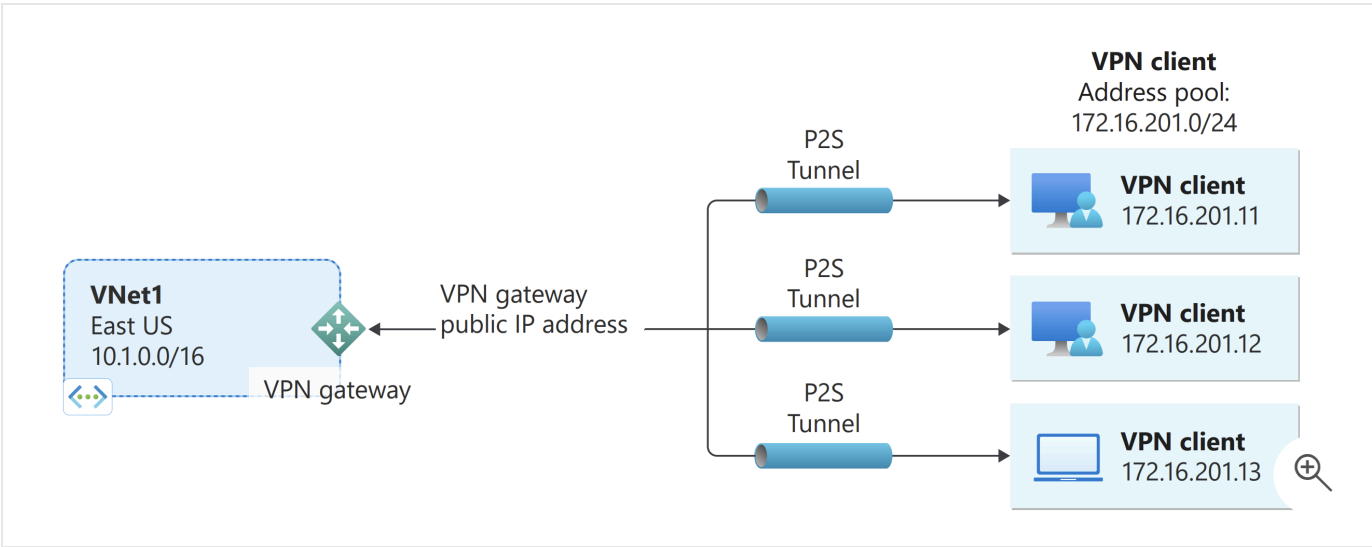
[Expand table](#)

Authentication method	Article
Pre-shared key	Portal PowerShell CLI
Certificate	Portal

Point-to-site VPN

A point-to-site (P2S) VPN gateway connection lets you create a secure connection to your virtual network from an individual client computer. A point-to-site connection is established by starting it from the client computer. This solution is useful for telecommuters who want to connect to Azure virtual networks from a remote location, such as from home or a conference. Point-to-site VPN is also a useful solution to use instead of site-to-site VPN when you have only a few clients that need to connect to a virtual network.

Unlike site-to-site connections, point-to-site connections don't require an on-premises public-facing IP address or a VPN device. Point-to-site connections can be used with site-to-site connections through the same VPN gateway, as long as all the configuration requirements for both connections are compatible. For more information about point-to-site connections, see [About point-to-site VPN](#).



Deployment methods for P2S

[Expand table](#)

Authentication method	Article
Certificate	Portal PowerShell
Microsoft Entra ID	Microsoft-registered client app ID Manually registered client app ID
RADIUS	Portal PowerShell

P2S VPN client configuration

[Expand table](#)

Authentication method	Tunnel type	Client OS	VPN client
Certificate	IKEv2, SSTP	Windows	Native VPN client
	IKEv2	macOS	Native VPN client
	IKEv2	Linux	strongSwan
	OpenVPN	Windows	Azure VPN client OpenVPN client version 2.x OpenVPN client version 3.x
	OpenVPN	macOS	OpenVPN client

Authentication method	Tunnel type	Client OS	VPN client
Microsoft Entra ID	OpenVPN	iOS	OpenVPN client
	OpenVPN	Linux	Azure VPN Client OpenVPN client
	OpenVPN	Windows	Azure VPN client
	OpenVPN	macOS	Azure VPN Client
	OpenVPN	Linux	Azure VPN Client

VNet-to-VNet connections (IPsec/IKE VPN tunnel)

Connecting a virtual network to another virtual network (VNet-to-VNet) is similar to connecting a virtual network to an on-premises site location. Both connectivity types use a VPN gateway to provide a secure tunnel using IPsec/IKE. You can even combine VNet-to-VNet communication with multi-site connection configurations. This lets you establish network topologies that combine cross-premises connectivity with inter-virtual network connectivity.

The virtual networks you connect can be:

- in the same or different regions
- in the same or different subscriptions



Deployment methods for VNet-to-VNet

[Expand table](#)

Connection	Article
VNet-to-VNet	Portal + PowerShell CLI

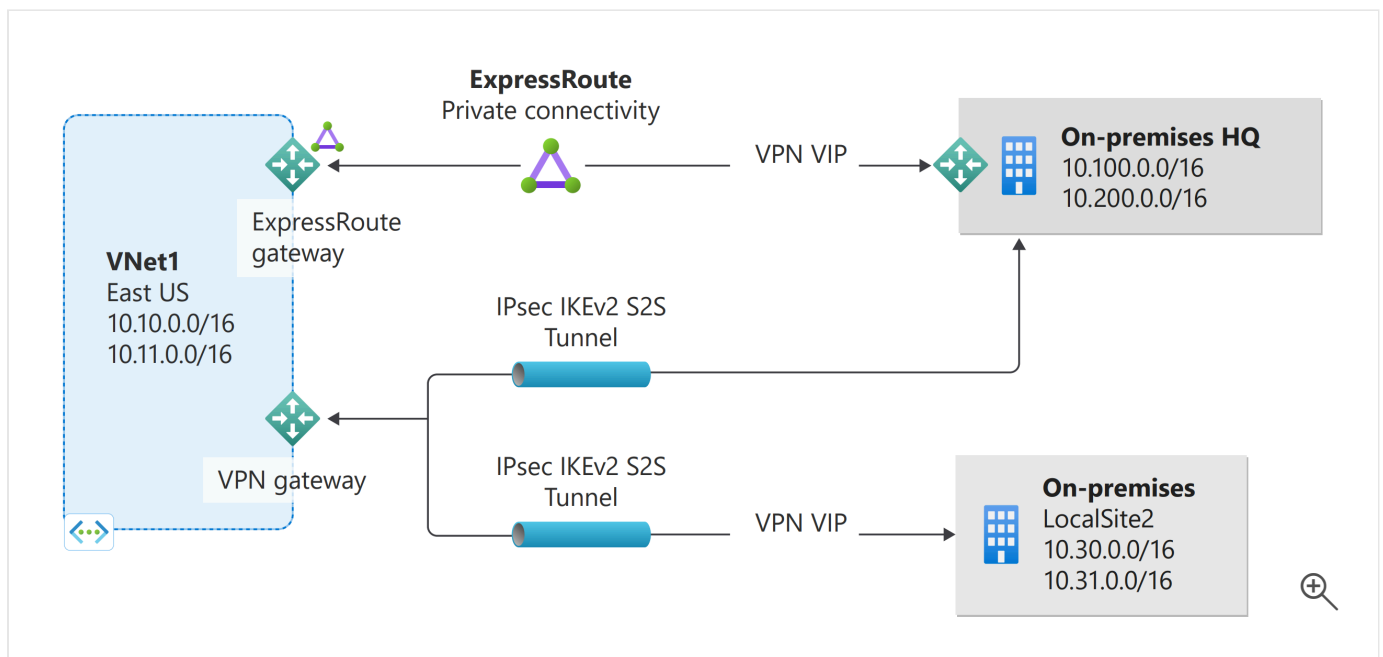
(+) Denotes this deployment method is available only for VNets in the same subscription.

In some cases, you might want to use virtual network peering instead of VNet-to-VNet to connect your virtual networks. Virtual network peering doesn't use a virtual network gateway. For more information, see [Virtual network peering](#).

Site-to-site and ExpressRoute coexisting connections

[ExpressRoute](#) is a direct, private connection from your WAN (not over the public Internet) to Microsoft Services, including Azure. Site-to-site VPN traffic travels encrypted over the public Internet. Being able to configure site-to-site VPN and ExpressRoute connections for the same virtual network has several advantages.

You can configure a site-to-site VPN as a secure failover path for ExpressRoute, or use site-to-site VPNs to connect to sites that aren't part of your network, but that are connected through ExpressRoute. Notice that this configuration requires two virtual network gateways for the same virtual network, one using the gateway type *Vpn*, and the other using the gateway type *ExpressRoute*.



Deployment methods for S2S and ExpressRoute coexisting connections

[Expand table](#)

Connection	Article
Coexisting connections	Portal PowerShell

Highly available connections

For planning and designing highly available connections, including active-active mode configurations, see [Design highly available gateway connectivity for cross-premises and VNet-to-VNet connections](#).

Next steps

- View the [VPN Gateway FAQ](#) for additional information.
- Learn more about [VPN Gateway configuration settings](#).
- For VPN Gateway BGP considerations, see [About BGP](#).
- For information about virtual network peering, see [Virtual network peering](#).
- View the [Subscription and service limits](#).
- Learn about some of the other key [networking capabilities](#) of Azure.

Last updated on 01/16/2025

Azure network security groups overview

You can use an Azure network security group to filter network traffic between Azure resources in Azure virtual networks. A network security group contains [security rules](#) that allow or deny inbound network traffic to, or outbound network traffic from, several types of Azure resources.

This article explains the properties of a network security group rule and the [default security rules](#) applied by Azure. It also describes how to modify rule properties to create an [augmented security rule](#).

Security rules

A network security group contains network security rules as desired, within Azure subscription [limits](#). Each rule specifies the following properties:

 Expand table

Property	Explanation
Name	A unique name within the network security group. The name can be up to 80 characters long. It must begin with a word character, and it must end with a word character or with <code>_</code> . The name can contain word characters, <code>.</code> , <code>-</code> , or <code>_</code> .
Priority	A number between 100 and 4096. Rules are processed in priority order, with lower numbers processed before higher numbers because lower numbers have higher priority. Once traffic matches a rule, processing stops. As a result, any rules that exist with lower priorities (higher numbers) that have the same attributes as rules with higher priorities aren't processed. Azure default security rules are given the lowest priority (highest number) to ensure your custom rules are always processed first.
Source or destination	You can specify Any, an individual IP address, a CIDR block (for example, 10.0.0.0/24), a service tag, or an application security group. To specify a particular Azure resource, use the private IP address assigned to the resource. For inbound traffic, network security groups process traffic after Azure translates public IP addresses to private IP addresses. For outbound traffic, network security groups process traffic before translating private IP addresses to public IP addresses. Enter a range, service tag, or application security group to reduce the number of security rules needed. Augmented security rules allow specifying multiple individual IP addresses and ranges in a single rule. However, you can't specify multiple service tags or application groups in a single rule. Augmented security rules are only available in network security groups created through the Resource Manager deployment model. In the classic deployment model, multiple IP addresses and ranges can't be specified in a single rule. For example, if the source is subnet 10.0.1.0/24 (where VM1 is located) and the destination is

Property	Explanation
	subnet 10.0.2.0/24 (where VM2 is located), the network security group filters traffic for VM2. This behavior occurs because the NSG is associated with VM2's network interface.
Protocol	TCP, UDP, ICMP, ESP, AH, or Any. The ESP and AH protocols aren't currently available via the Azure portal but can be used via ARM templates.
Direction	Whether the rule applies to inbound or outbound traffic.
Port range	You can specify an individual port or ranges of ports. For example, you could specify 80 or 10000-10005; or for a mix of individual ports and ranges, you can separate them with commas, such as 80, 10000-10005. Specifying ranges and comma separation empowers you to create fewer security rules. Augmented security rules can only be created in network security groups created through the Resource Manager deployment model. You can't specify multiple ports or port ranges in the same security rule in network security groups created through the classic deployment model.
Action	Allow or deny your specified traffic.

Security rules are evaluated and applied based on the five-tuple information of source, source port, destination, destination port, and protocol. You can't create two security rules with the same priority and direction. Two security rules with the same priority and direction can introduce a conflict in how the system processes traffic. A flow record is created for existing connections. Communication is allowed or denied based on the connection state of the flow record. The flow record allows a network security group to be stateful. If you specify an outbound security rule to any address over port 80, for example, it's not necessary to specify an inbound security rule for the response to the outbound traffic. You only need to specify an inbound security rule if communication is initiated externally. The opposite is true. If inbound traffic is allowed over a port, it's not necessary to specify an outbound security rule to respond to traffic over the port.

When you remove a security rule that allowed a connection, existing connections remain uninterrupted. Network security group rules only affect new connections. New or updated rules in a network security group apply exclusively to new connections, leaving existing connections unaffected by the changes. For example, if you have an active SSH session to a virtual machine and then remove the security rule allowing that SSH traffic, your current SSH session remains connected and functional. However, if you try to establish a new SSH connection after the security rule removal, that new connection attempt will be blocked.

There are limits to the number of security rules you can create in a network security group and other properties of the network security group. For details, see [Azure limits](#).

Default security rules

Azure creates the following default rules in each network security group that you create:

Inbound

AllowVNetInBound

[Expand table](#)

Priority	Source	Source ports	Destination	Destination ports	Protocol	Access
65000	VirtualNetwork	0-65535	VirtualNetwork	0-65535	Any	Allow

AllowAzureLoadBalancerInBound

[Expand table](#)

Priority	Source	Source ports	Destination	Destination ports	Protocol	Access
65001	AzureLoadBalancer	0-65535	0.0.0.0/0	0-65535	Any	Allow

DenyAllInbound

[Expand table](#)

Priority	Source	Source ports	Destination	Destination ports	Protocol	Access
65500	0.0.0.0/0	0-65535	0.0.0.0/0	0-65535	Any	Deny

Outbound

AllowVnetOutBound

[Expand table](#)

Priority	Source	Source ports	Destination	Destination ports	Protocol	Access
65000	VirtualNetwork	0-65535	VirtualNetwork	0-65535	Any	Allow

AllowInternetOutBound

[Expand table](#)

Priority	Source	Source ports	Destination	Destination ports	Protocol	Access
65001	0.0.0.0/0	0-65535	Internet	0-65535	Any	Allow

DenyAllOutBound

[Expand table](#)

Priority	Source	Source ports	Destination	Destination ports	Protocol	Access
65500	0.0.0.0/0	0-65535	0.0.0.0/0	0-65535	Any	Deny

In the **Source** and **Destination** columns, *VirtualNetwork*, *AzureLoadBalancer*, and *Internet* are [service tags](#) rather than IP addresses. In the **Protocol** column, *Any* encompasses TCP, UDP, and ICMP. When creating a rule, you can specify TCP, UDP, ICMP, or Any. *0.0.0.0/0* in the **Source** and **Destination** columns represents all IP addresses. Clients like the Azure portal, Azure CLI, or PowerShell can use *** or *Any* for this expression.

You can't remove the default rules, but you can override them by creating rules with higher priorities.

Augmented security rules

Augmented security rules simplify the security definition for virtual networks, allowing you to define larger and complex network security policies with fewer rules. You can combine multiple ports and multiple explicit IP addresses and ranges into a single, easily understood security rule. Use augmented rules in the source, destination, and port fields of a rule. To simplify maintenance of your security rule definition, combine augmented security rules with [service tags](#) or [application security groups](#). There are limits to the number of addresses, ranges, and ports that you can specify in a security rule. For details, see [Azure limits](#).

Service tags

A service tag represents a group of IP address prefixes from a given Azure service. It helps minimize the complexity of frequent updates on network security rules.

For more information, see [Azure service tags](#). For an example on how to use the Storage service tag to restrict network access, see [Restrict network access to PaaS resources](#).

Application security groups

Application security groups enable you to configure network security as a natural extension of an application's structure, allowing you to group virtual machines and define network security policies based on those groups. You can reuse your security policy at scale without manual maintenance of explicit IP addresses. To learn more, see [Application security groups](#).

Security admin rules

Security admin rules are global network security rules that enforce security policies onto virtual networks. Security admin rules originate from Azure Virtual Network Manager, a management service that enables network administrators to group, configure, deploy, and manage virtual networks globally across subscriptions.

Security admin rules always have a higher priority than network security group rules and thus are evaluated first. "Allow" security admin rules continue for evaluation by matching network security group rules. "Always allow" and "Deny" security admin rules, however, terminate traffic evaluation after the security admin rule is processed. "Always allow" security admin rules send traffic directly to the resource, bypassing potentially conflicting network security group rules. "Deny" security admin rules block the traffic without delivering it to the destination. These rules enforce baseline security policy without risk of network security group conflict, misconfiguration, or introduction of security gaps. These security admin rule action types can be useful for enforcing traffic delivery and preventing conflicting or unintended behavior by downstream network security group rules.

This behavior is important to understand, as traffic matching security admin rules of "Always allow" or "Deny" action types don't reach network security group rules for further evaluation. To learn more, see [Security admin rules](#).

Flow timeout

Important

Network security group (NSG) flow logs will be retired on September 30, 2027. After June 30, 2025, you'll no longer be able to create new NSG flow logs. We recommend [migrating](#) to [virtual network flow logs](#), which address the limitations of NSG flow logs. After the retirement date, traffic analytics enabled for NSG flow logs will no longer be supported, and existing NSG flow log resources in your subscriptions will be deleted. However, existing NSG flow log records won't be deleted from Azure Storage and will continue to follow their configured retention policies. For more information, see the [official announcement](#).

Flow timeout settings determine how long a flow record remains active before expiring. You can configure this setting using the Azure portal or through the command line. For more information, see [NSG flow logs overview](#).

Azure platform considerations

- **Virtual IP of the host node:** Basic infrastructure services like DHCP, DNS, IMDS, and health monitoring are provided through the virtualized host IP addresses 168.63.129.16 and 169.254.169.254. These IP addresses belong to Microsoft and are the only virtualized IP addresses used in all regions for this purpose. By default, these services aren't subject to the configured network security groups unless targeted by [service tags](#) specific to each service. To override this basic infrastructure communication, you can create a security rule to deny traffic by using the following service tags on your Network Security Group rules: AzurePlatformDNS, AzurePlatformIMDS, AzurePlatformLKM. Learn how to [diagnose network traffic filtering](#) and [diagnose network routing](#).
- **Licensing (Key Management Service):** Windows images running in virtual machines must be licensed. To ensure licensing, the system sends a request to the Key Management Service host servers that handle such queries. The request is made outbound through port 1688. For deployments using [default route 0.0.0.0/0](#) configuration, this platform rule is disabled.
- **Virtual machines in load-balanced pools:** The source port and address range applied are from the originating computer, not the load balancer. The destination port and address range are for the destination computer, not the load balancer.
- **Azure service instances:** Instances of several Azure services, such as HDInsight, Application Service Environments, and Virtual Machine Scale Sets, are deployed in virtual network subnets. See a complete list of [services you can deploy into virtual networks](#). Before applying a network security group to the subnet, familiarize yourself with the port requirements for each service. If you deny ports required by the service, the service doesn't function properly.
- **Sending outbound email:** Microsoft recommends that you utilize authenticated SMTP relay services (typically connected via TCP port 587, but often others, as well) to send email from Azure Virtual Machines. SMTP relay services specialize in sender reputation, to minimize the possibility that partner email providers reject messages. Such SMTP relay services include, but aren't limited to, Exchange Online Protection and SendGrid. Use of SMTP relay services is in no way restricted in Azure, regardless of your subscription type.

If you created your Azure subscription before November 15, 2017, in addition to being able to use SMTP relay services, you can send email directly over TCP port 25. If you created your subscription after November 15, 2017, you might not be able to send email directly over

port 25. The behavior of outbound communication over port 25 depends on the type of subscription you have, as follows:

- **Enterprise Agreement:** For VMs that are deployed in standard Enterprise Agreement subscriptions, the outbound SMTP connections on TCP port 25 aren't blocked. However, there's no guarantee that external domains accept the incoming emails from the VMs. If external domains reject or filter emails, contact the email service providers of the external domains to resolve the problems. These problems aren't covered by Azure support.

For Enterprise Dev/Test subscriptions, port 25 is blocked by default. It's possible to have this block removed. To request to have the block removed, go to the **Can't send email (SMTP-Port 25)** section of the **Diagnose and Solve** settings page for the Azure Virtual Network resource in the Azure portal and run the diagnostic. This procedure exempts the qualified enterprise dev/test subscriptions automatically.

After the subscription is exempted from this block and the VMs are stopped and restarted, all VMs in that subscription are exempted going forward. The exemption applies only to the subscription requested and only to VM traffic that routes directly to the internet.

- **Pay-as-you-go:** Outbound port 25 communication is blocked from all resources. No requests to remove the restriction can be made, because requests aren't granted. If you need to send email from your virtual machine, you have to use an SMTP relay service.
- **MSDN, Azure Pass, Azure in Open, Education, and Free trial:** Outbound port 25 communication is blocked from all resources. No requests to remove the restriction can be made, because requests aren't granted. If you need to send email from your virtual machine, you have to use an SMTP relay service.
- **Cloud service provider:** Outbound port 25 communication is blocked from all resources. No requests to remove the restriction can be made, because requests aren't granted. If you need to send email from your virtual machine, you have to use an SMTP relay service.

Next steps

- Learn which Azure resources you can deploy into a virtual network. See [Virtual network integration for Azure services](#) to understand how network security groups can be associated with them.

- To learn how network security groups evaluate traffic, see [How network security groups work](#).
- Create a network security group by following this quick [tutorial](#).
- If you're familiar with network security groups and need to manage them, see [Manage a network security group](#).
- If you're having communication problems and need to troubleshoot network security groups, see [Diagnose a virtual machine network traffic filter problem](#).
- Learn how to enable [virtual network flow logs](#) to analyze network traffic flowing through a virtual network that might match an associated network security group.

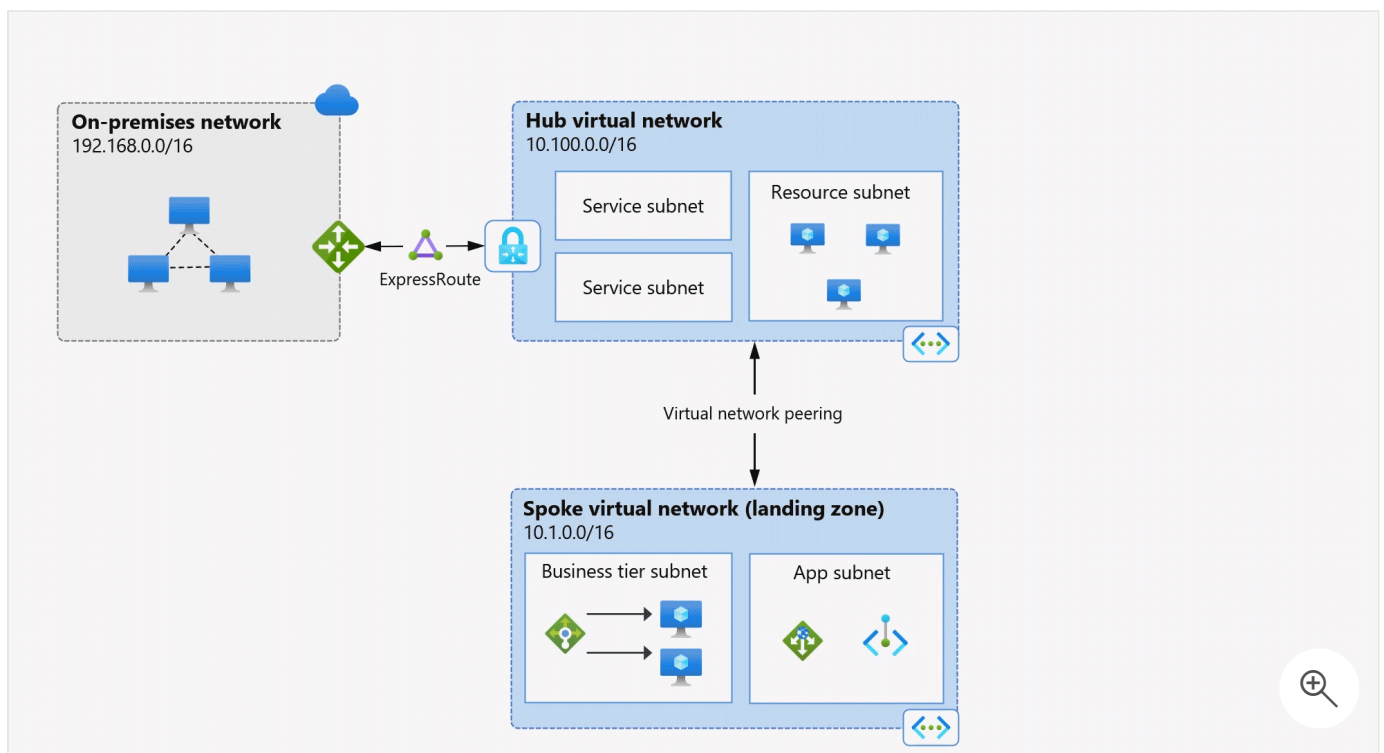
Last updated on 07/15/2025

Azure network foundation services overview

Azure network foundation services provide core connectivity for your resources in Azure.

Network foundation services include **Azure Virtual Network**, **Azure Private Link**, and **Azure DNS**. Together, these core services build upon each other to provide the foundation for your Azure network.

The following diagram is an example of how these services can be used together in a basic Azure network.



This article provides a summary of each of these Azure foundational services, and illustrates how they work together. Links are also provided to more detailed guidance for each foundational service.

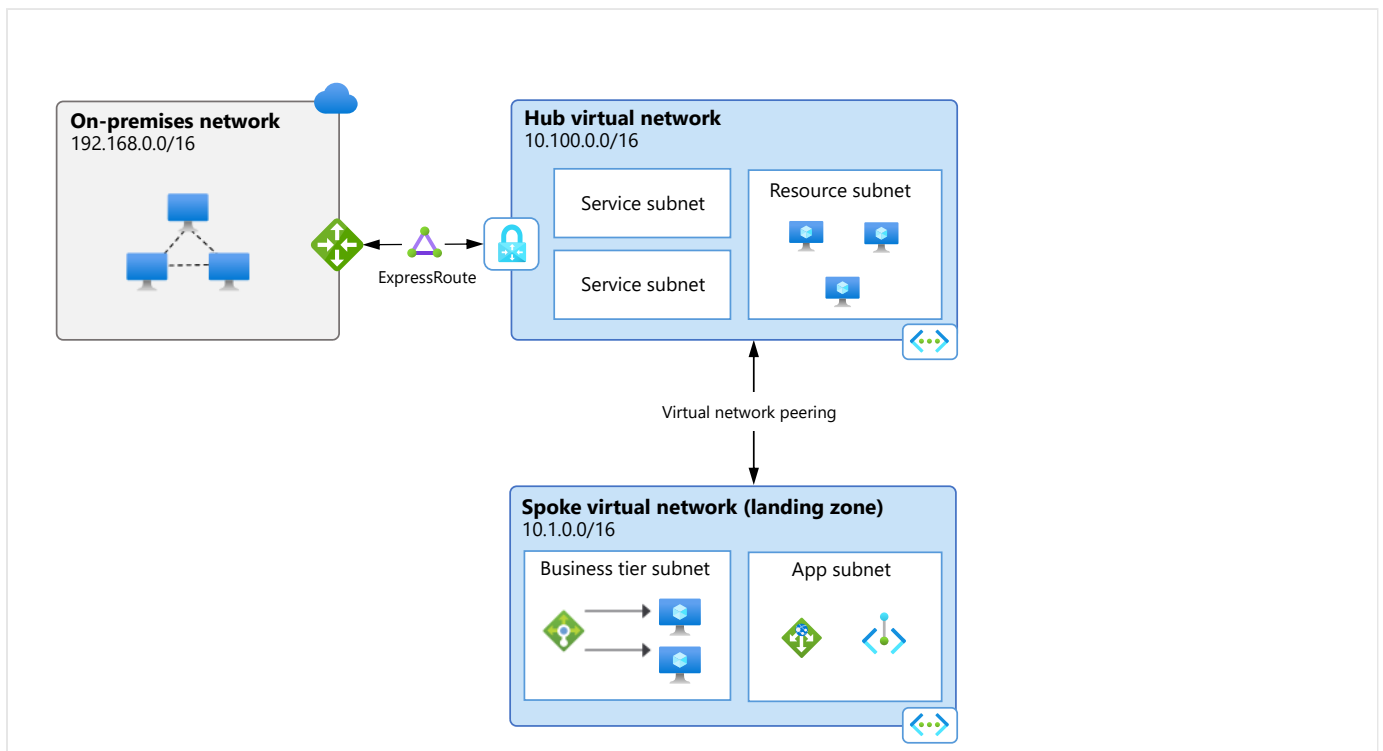
Azure Virtual Network

[Azure Virtual Network](#) enables you to create private networks in the cloud, securely connecting Azure resources, the Internet, and on-premises networks.

Two virtual networks are provisioned in the following example:

- The hub virtual network is used to deploy Azure services and provide access to data resources. The hub is optionally connected to an on-premises network.

- The hub peers with a spoke network containing a business tier subnet with virtual machines to process user interactions, and an application subnet to handle data storage and transactions.



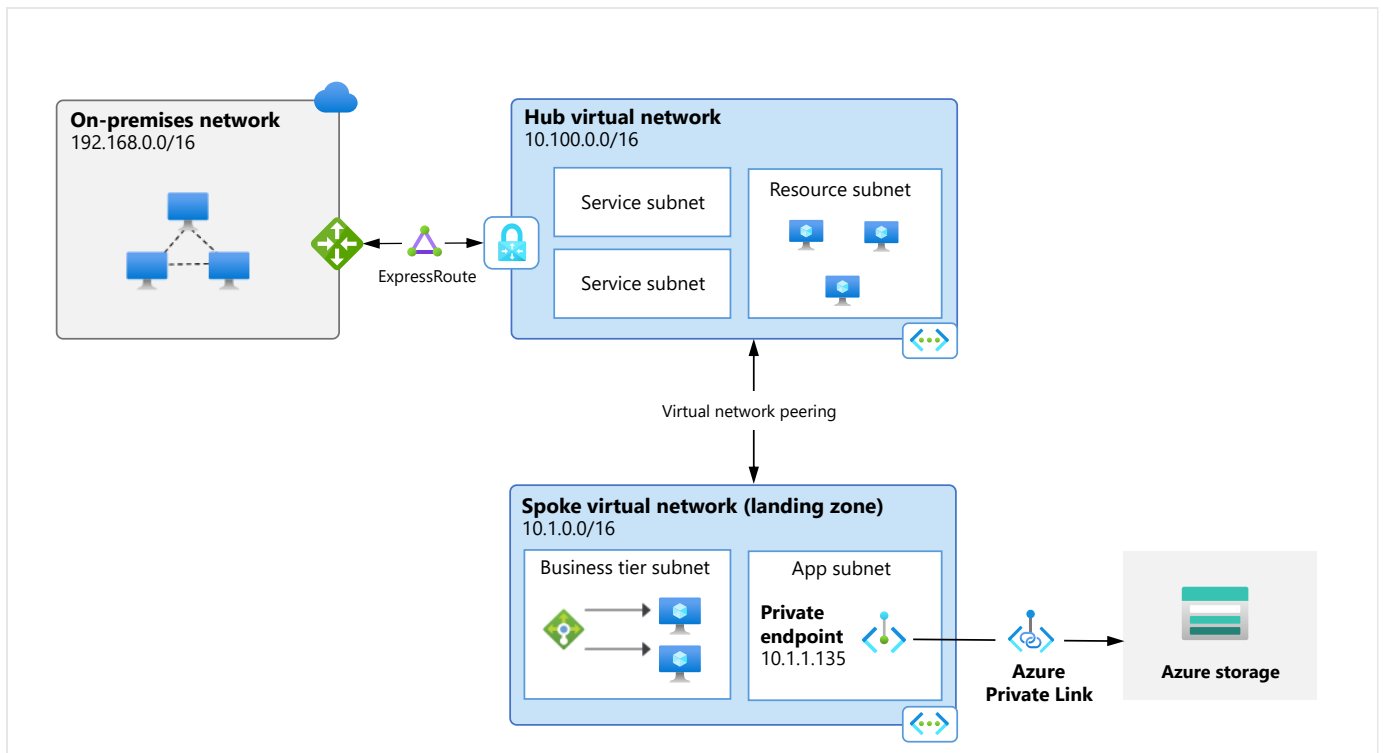
For more information about designing virtual networks, see [Plan virtual networks](#). To create a virtual network, see [Use the Azure portal to create a virtual network](#).

Azure Private Link

[Azure Private Link](#) enables secure, private connectivity from your virtual network to services that don't traverse the public Internet.

In the following figure, a **private endpoint** is provisioned in the app subnet of the spoke virtual network described previously. A private endpoint is a private IP address (10.1.1.135 in this example) associated with a service powered by Azure Private Link.

Private endpoints securely connect services within virtual networks.



Note

Private endpoints offer DNS integration options during creation. You can choose to integrate with a private DNS zone, and this configuration remains flexible - you can add, remove, or modify it after deployment. The example demonstrates selecting private DNS zone integration, which provides a straightforward DNS setup ideal for virtual network workloads without an Azure DNS Private Resolver. For more information, see [Azure Private Endpoint DNS integration](#).

For an overview of private link and private endpoint, see [What is Azure Private Link service](#) and [What is a private endpoint](#). To create a private endpoint, see [Create a private endpoint](#).

Azure DNS

Azure DNS provides cloud-based public and private domain name hosting and resolution. It includes three services that provide public or private DNS resolution and hosting, and one load balancing service:

- **Azure Public DNS** provides high-availability hosting for public DNS domains.
- **Azure Private DNS** is a DNS naming and resolution service for virtual networks and the private services hosted inside those networks.
- **Azure DNS Private Resolver** is a fully managed high availability DNS service that enables you to query private DNS zones from an on-premises environment and vice versa without

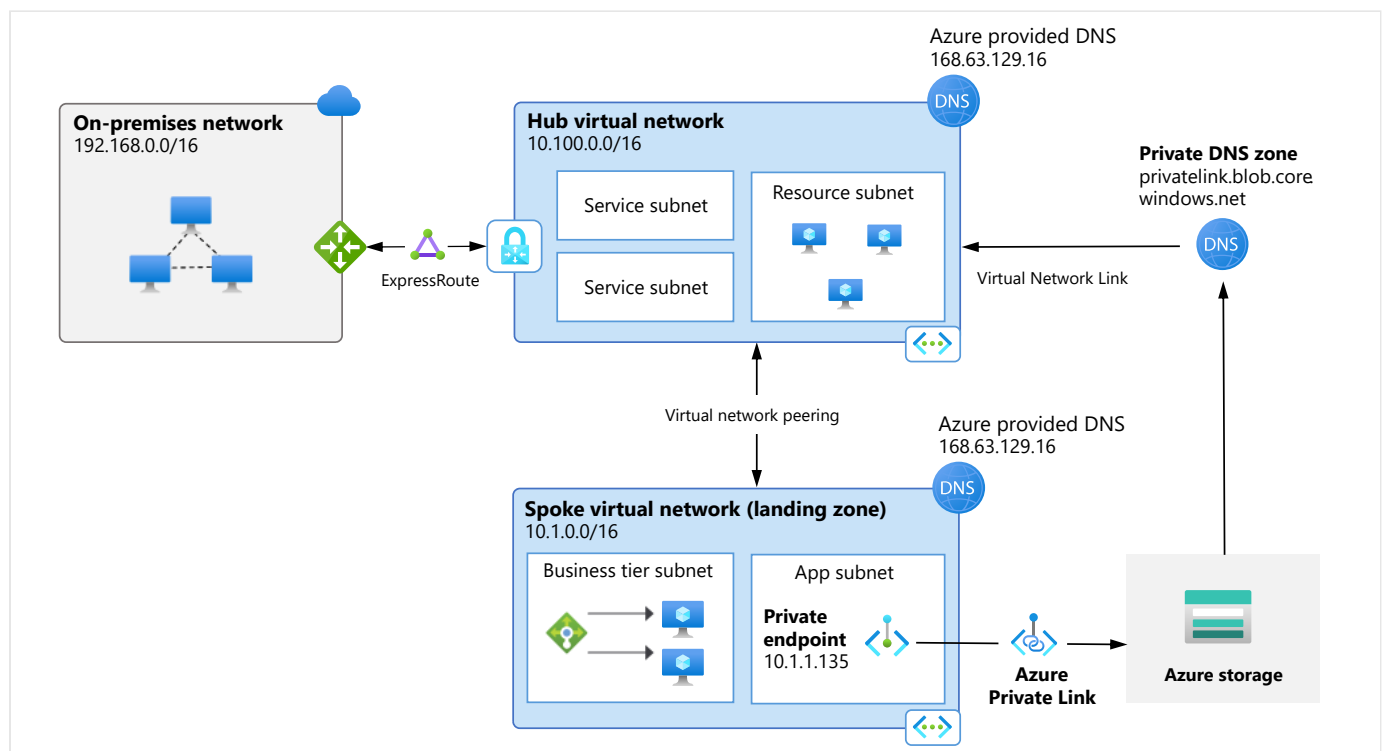
deploying VM based DNS servers.

- **Azure Traffic Manager:** A DNS-based traffic load balancer that can distribute traffic to public facing applications across Azure regions.

Azure DNS also provides internal DNS resolution for both private and public (internet) resources from within virtual networks. By default, virtual networks are configured to resolve DNS records using Azure-provided DNS at [168.63.129.16](#).

In the following example, the private endpoint shown in the previous figure is associated with the private DNS zone **privatelink.blob.core.windows.net**. The private DNS zone name is dependent on the type of private endpoint resource. In this example, the resource is blob storage. For more information about private DNS zones and private endpoints, see [Azure Private Endpoint private DNS zone values](#).

This zone is also configured with a virtual network link to the hub virtual network, enabling all resources in the hub network to resolve the zone using Azure-provided DNS (168.63.129.16) and providing access to the private endpoint using its fully qualified DNS name (FQDN).



By default, private endpoints can only be resolved from within Azure. To resolve the private-linked storage account from on-premises, or to resolve on-premises resources from within Azure, you can configure a **DNS private resolver** in the hub virtual network (not shown).

For information about private endpoint scenarios with an Azure DNS Private Resolver, see [private endpoint DNS configuration scenarios](#).

For more information about configuring a DNS private resolver, see [Resolve Azure and on-premises domains](#).

Azure portal experience

The Azure portal provides a centralized experience for [getting started with network foundation services](#). Information and links are provided to help you create an isolated network, manage network services, secure access to resources, manage hybrid name resolution, and troubleshoot network issues.

The screenshot shows the 'Network foundation | Overview' page in the Azure portal. The left-hand navigation pane is expanded, showing categories like 'Virtual network', 'Private Link', 'DNS', and 'Monitoring and management'. The main content area is titled 'Build your network foundation' and includes sections for 'Services to create your virtual network' (Virtual network, Private Link, DNS) and 'Common network foundation scenarios' (Create isolated network, Virtual network, Virtual network managers, Virtual machine). Each scenario includes a brief description and a 'Create' button. A diagram on the right illustrates a network architecture with a central cloud icon connected to two server icons.

Resource links are also provided in the left-hand service tree to help you understand, create, and view supporting components of the network foundation services.

Learn more

- [Load balancing and content delivery](#)
- [Azure hybrid connectivity](#)
- [Azure network security](#)

- [Azure network monitoring and management](#)
 - [Azure Networking Fundamentals](#)
 - [Azure networking](#)
-

Last updated on 07/26/2025

Azure networking services overview

The networking services in Azure provide various networking capabilities that can be used together or separately. Select each of the following networking scenarios to learn more about them:

- **Networking foundation:** [Azure networking foundation services](#) provide core connectivity for your resources in Azure - Virtual Network (VNet), Private Link, Azure DNS, Azure Bastion, Route Server, NAT Gateway, and Traffic Manager.
- **Load balancing and content delivery:** [Azure load balancing and content delivery services](#) allow for management, distribution, and optimization of your applications and workloads - Load balancer, Application Gateway, and Azure Front Door.
- **Hybrid connectivity:** [Azure hybrid connectivity services](#) secure communication to and from your resources in Azure - VPN Gateway, ExpressRoute, Virtual WAN, and Peering Service.
- **Network security:** [Azure network security services](#) protect your web applications and IaaS services from DDoS attacks and malicious actors - Firewall Manager, Firewall, Web Application Firewall, and DDoS Protection.
- **Network Management and monitoring:** [Azure network management and monitoring services](#) provide tools to manage and monitor your network resources - Network Watcher, Azure Monitor, and Azure Virtual Network Manager.

Networking foundation

This section describes services that provide the building blocks for designing and architecting a network environment in Azure - Virtual Network (VNet), Private Link, Azure DNS, Azure Bastion, Route Server, NAT Gateway, and Traffic Manager.

Virtual network

[Azure Virtual Network \(VNet\)](#) is the fundamental building block for your private network in Azure. You can use VNets to:

- **Communicate between Azure resources:** You can deploy virtual machines, and several other types of Azure resources to a virtual network, such as Azure App Service Environments, the Azure Kubernetes Service (AKS), and Azure Virtual Machine Scale Sets. To view a complete list of Azure resources that you can deploy into a virtual network, see [Virtual network service integration](#).

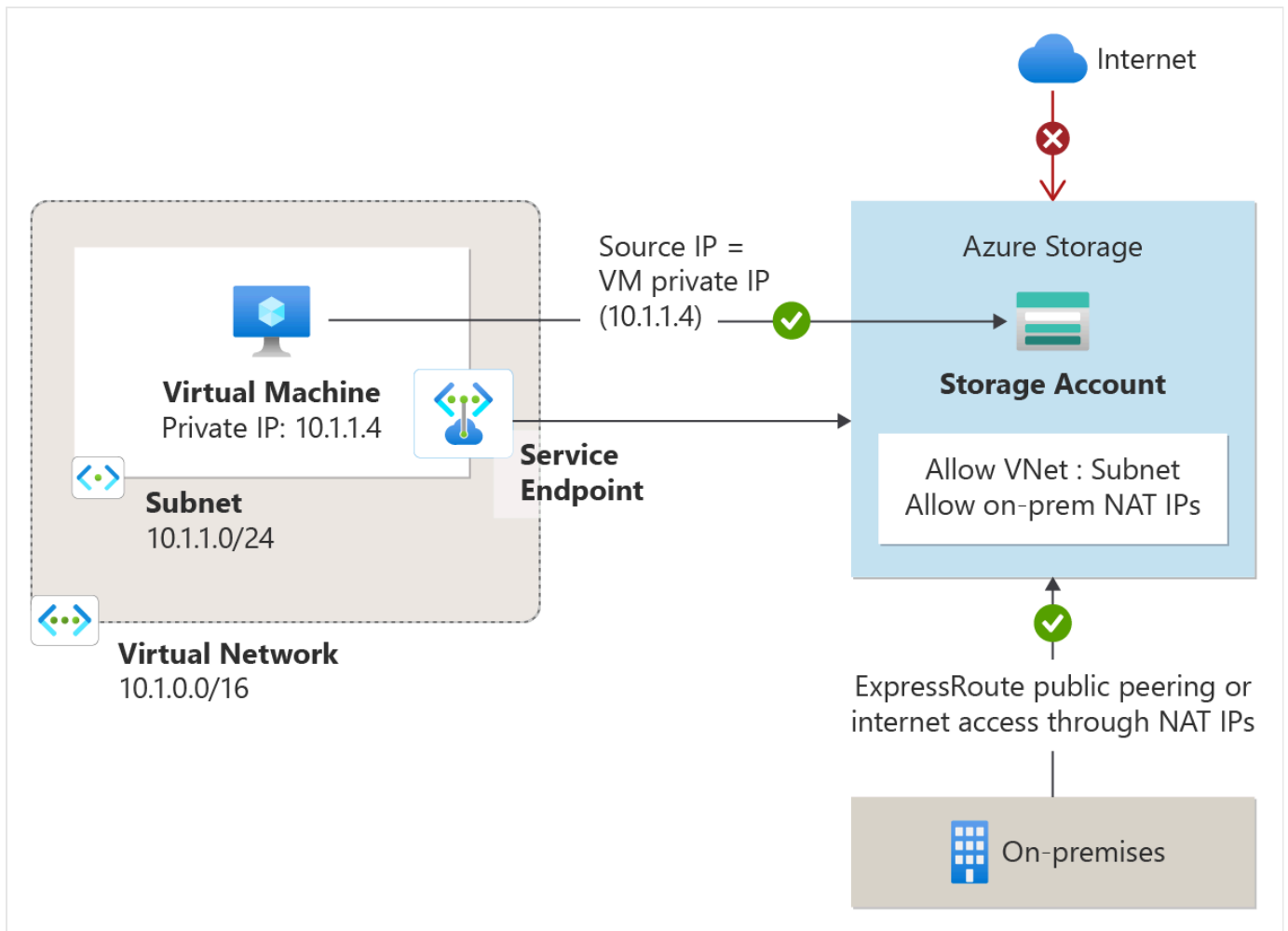
- **Communicate between each other:** You can connect virtual networks to each other, enabling resources in either virtual network to communicate with each other, using virtual network peering or Azure Virtual Network Manager. The virtual networks you connect can be in the same, or different, Azure regions. For more information, see [Virtual network peering](#) and [Azure Virtual Network Manager](#).
- **Communicate to the internet:** All resources in a virtual network can communicate outbound to the internet, by default. You can communicate inbound to a resource by assigning a public IP address or a public Load Balancer. You can also use [Public IP addresses](#) or public [Load Balancer](#) to manage your outbound connections.
- **Communicate with on-premises networks:** You can connect your on-premises computers and networks to a virtual network using [VPN Gateway](#) or [ExpressRoute](#).
- **Encrypt traffic between resources:** You can use [Virtual network encryption](#) to encrypt traffic between resources in a virtual network.

Network security groups

You can filter network traffic to and from Azure resources in an Azure virtual network with a network security group. For more information, see [Network security groups](#).

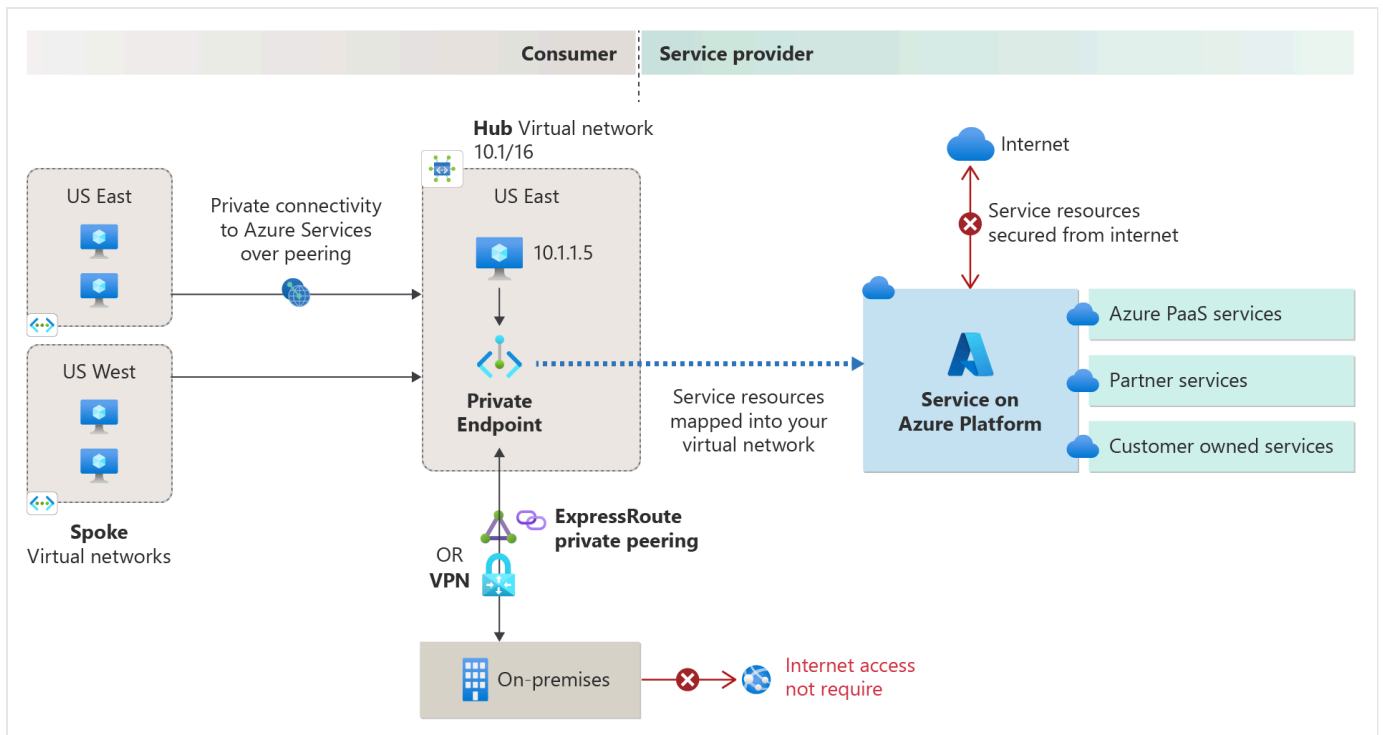
Service endpoints

[Virtual Network \(VNet\) service endpoints](#) extend your virtual network private address space and the identity of your virtual network to the Azure services, over a direct connection. Endpoints allow you to secure your critical Azure service resources to only your virtual networks. Traffic from your virtual network to the Azure service always remains on the Microsoft Azure backbone network.



▸ Azure Private Link

[Azure Private Link](#) enables you to access Azure PaaS Services (for example, Azure Storage and SQL Database) and Azure hosted customer-owned/partner services over a private endpoint in your virtual network. Traffic between your virtual network and the service travels through the Microsoft backbone network. Exposing your service to the public internet is no longer necessary. You can create your own private link service in your virtual network and deliver it to your customers.



Azure DNS

Azure DNS provides DNS hosting and resolution using the Microsoft Azure infrastructure. Azure DNS consists of three services:

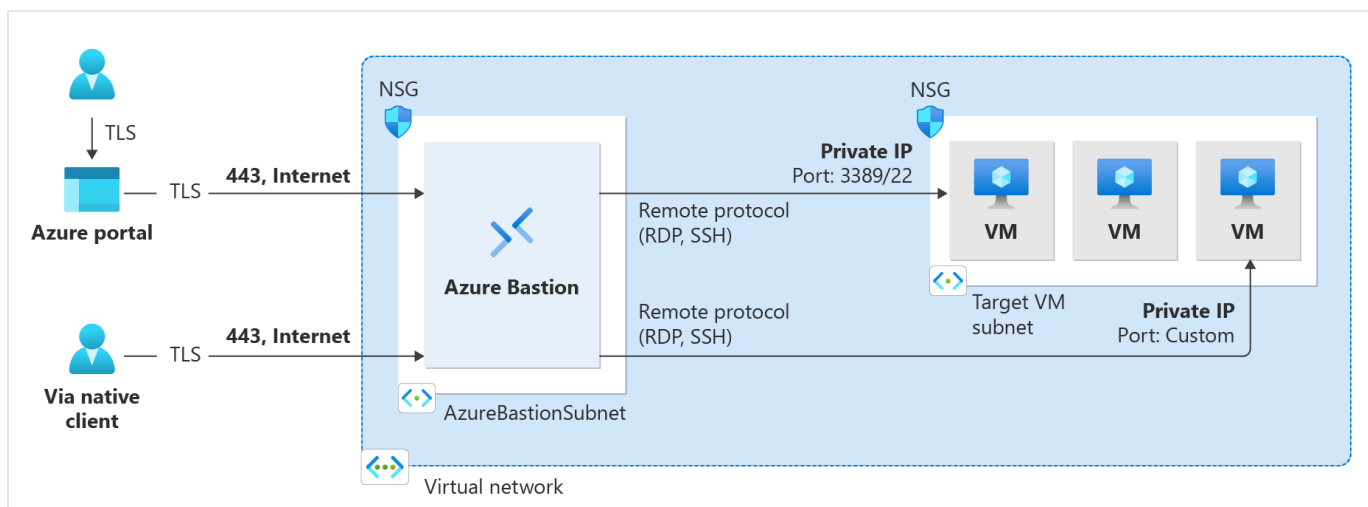
- **Azure Public DNS** is a hosting service for DNS domains. By hosting your domains in Azure, you can manage your DNS records by using the same credentials, APIs, tools, and billing as your other Azure services.
- **Azure Private DNS** is a DNS service for your virtual networks. Azure Private DNS manages and resolves domain names in the virtual network without the need to configure a custom DNS solution.
- **Azure DNS Private Resolver** is a service that enables you to query Azure DNS private zones from an on-premises environment and vice versa without deploying VM based DNS servers.

Using Azure DNS, you can host and resolve public domains, manage DNS resolution in your virtual networks, and enable name resolution between Azure and your on-premises resources.

Azure Bastion

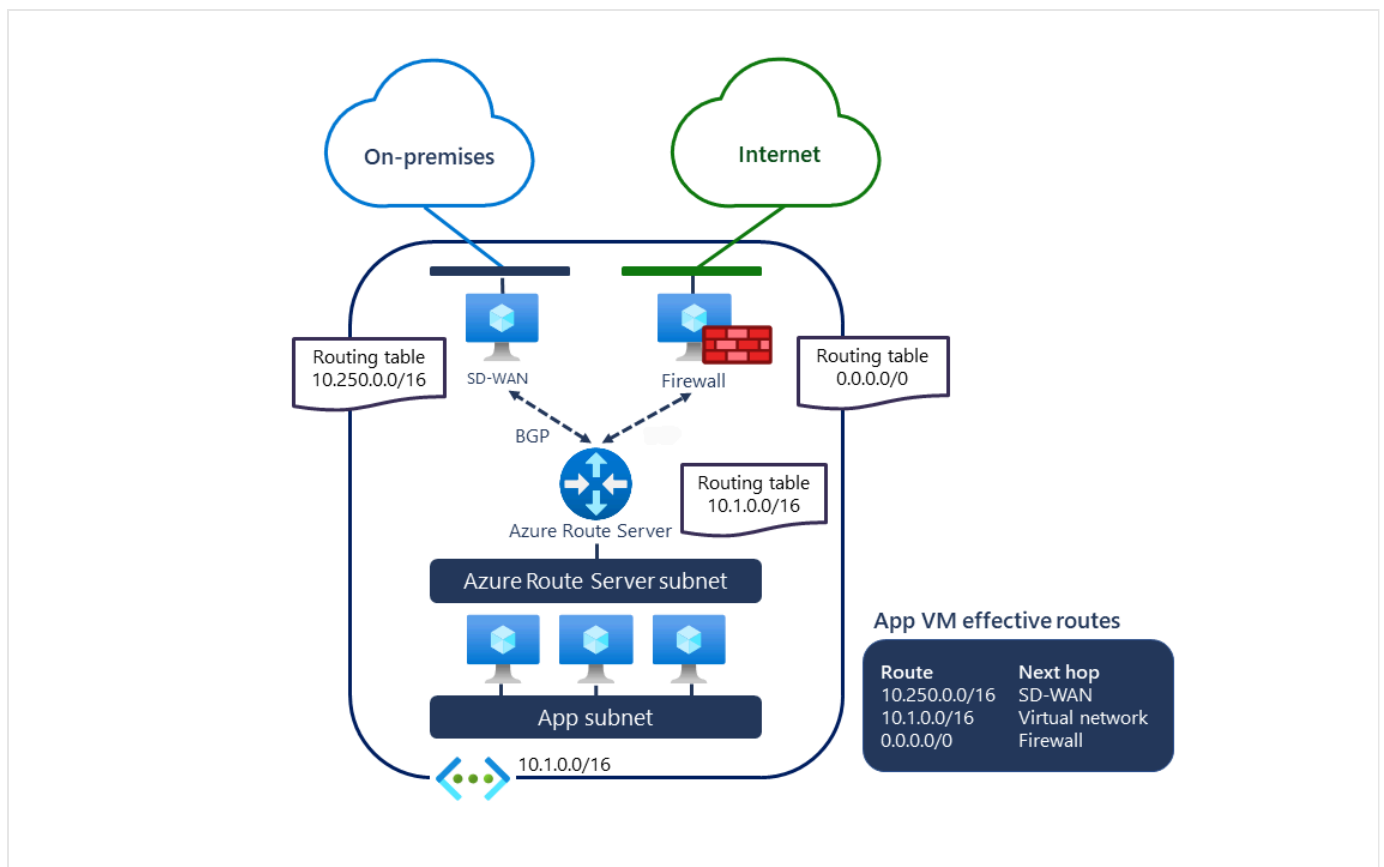
Azure Bastion is a service that you can deploy in a virtual network to allow you to connect to a virtual machine using your browser and the Azure portal. You can also connect using the native SSH or RDP client already installed on your local computer. The Azure Bastion service is a fully platform-managed PaaS service that you deploy inside your virtual network. It provides secure and seamless RDP/SSH connectivity to your virtual machines directly from the Azure portal over

TLS. When you connect via Azure Bastion, your virtual machines don't need a public IP address, agent, or special client software. There are various different SKU/tiers available for Azure Bastion. The tier you select affects the features that are available. For more information, see [About Bastion configuration settings](#).



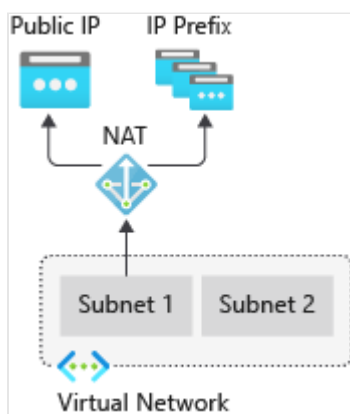
Azure Route Server

[Azure Route Server](#) simplifies dynamic routing between your network virtual appliance (NVA) and your virtual network. It allows you to exchange routing information directly through Border Gateway Protocol (BGP) routing protocol between any NVA that supports the BGP routing protocol and the Azure Software Defined Network (SDN) in the Azure Virtual Network (VNet) without the need to manually configure or maintain route tables.



NAT Gateway

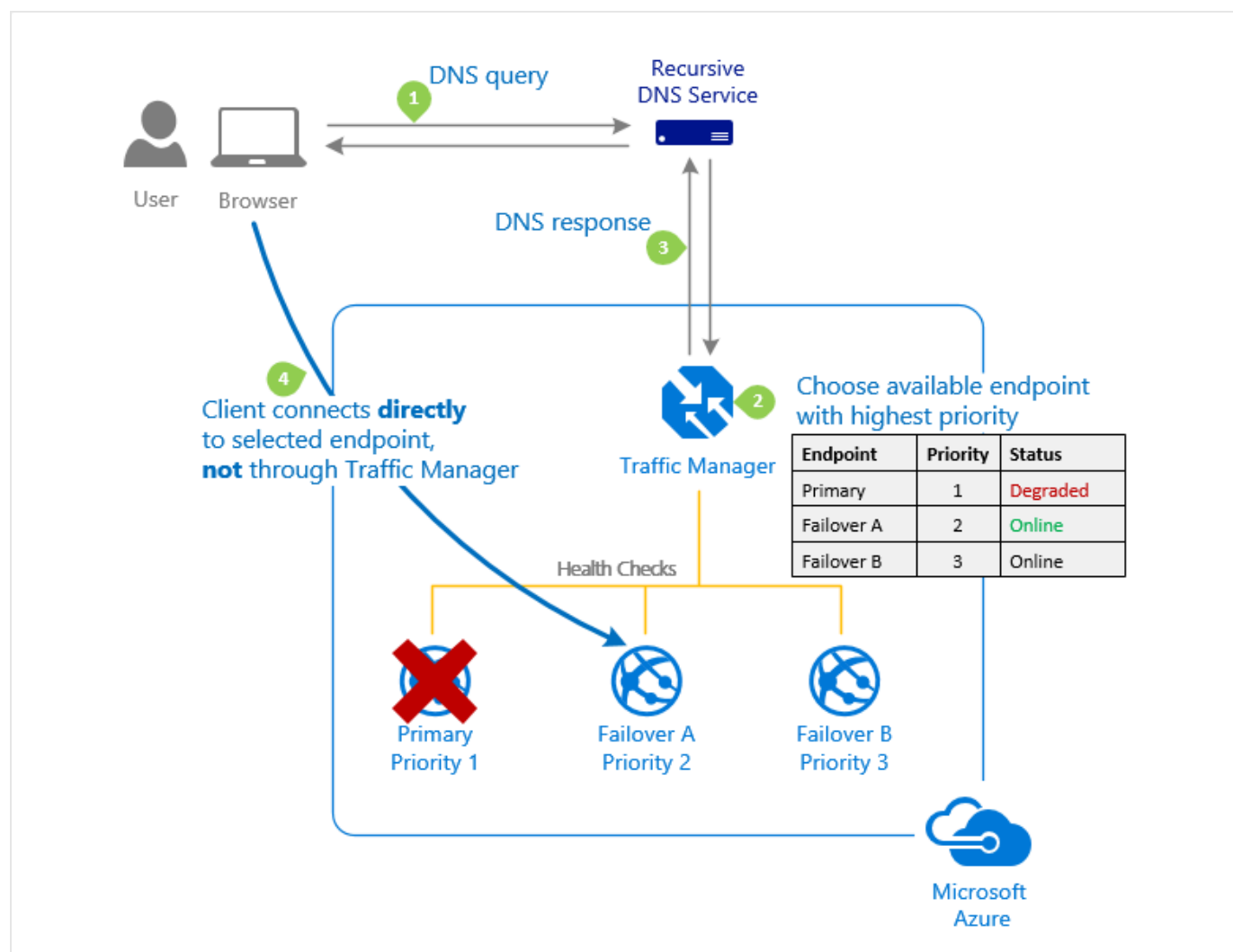
NAT Gateway simplifies outbound-only Internet connectivity for virtual networks. When configured on a subnet, all outbound connectivity uses your specified static public IP addresses. Outbound connectivity is possible without load balancer or public IP addresses directly attached to virtual machines. For more information, see [What is Azure NAT gateway?](#)



Traffic Manager

[Azure Traffic Manager](#) is a DNS-based traffic load balancer that enables you to distribute traffic optimally to services across global Azure regions, while providing high availability and responsiveness. Traffic Manager provides a range of traffic-routing methods to distribute traffic such as priority, weighted, performance, geographic, multi-value, or subnet.

The following diagram shows endpoint priority-based routing with Traffic Manager:



For more information about Traffic Manager, see [What is Azure Traffic Manager?](#)

Load balancing and content delivery

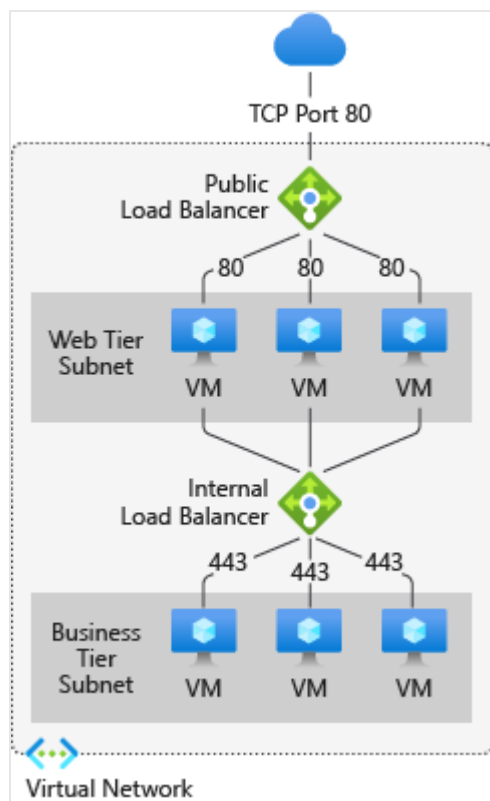
This section describes networking services in Azure that help deliver applications and workloads - Load Balancer, Application Gateway, and Azure Front Door Service.

Load Balancer

[Azure Load Balancer](#) provides high-performance, low-latency Layer 4 load-balancing for all UDP and TCP protocols. It manages inbound and outbound connections. You can configure public and internal load-balanced endpoints. You can define rules to map inbound connections to back-end pool destinations by using TCP and HTTP health-probing options to manage service availability.

Azure Load Balancer is available in Standard, Regional, and Gateway SKUs.

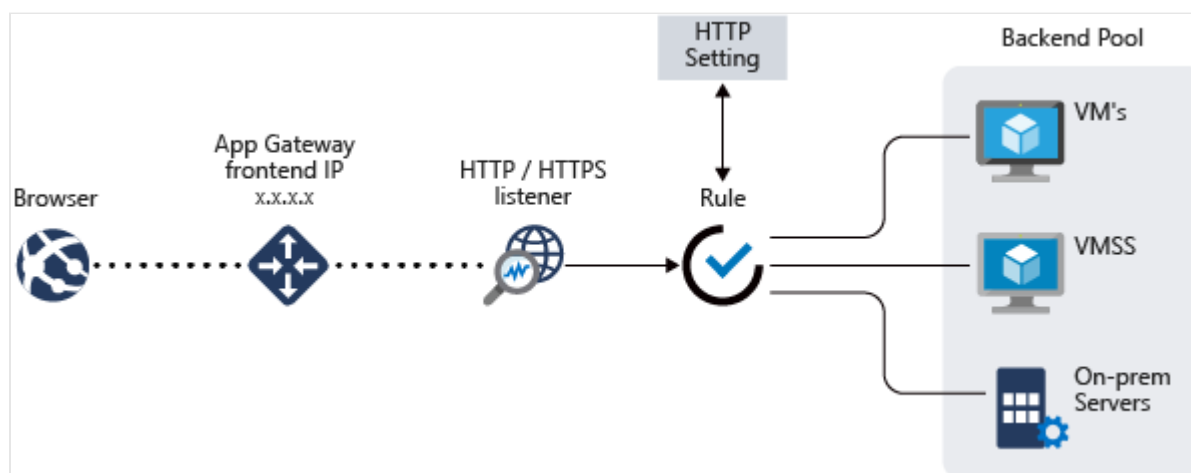
The following picture shows an Internet-facing multi-tier application that utilizes both external and internal load balancers:



Application Gateway

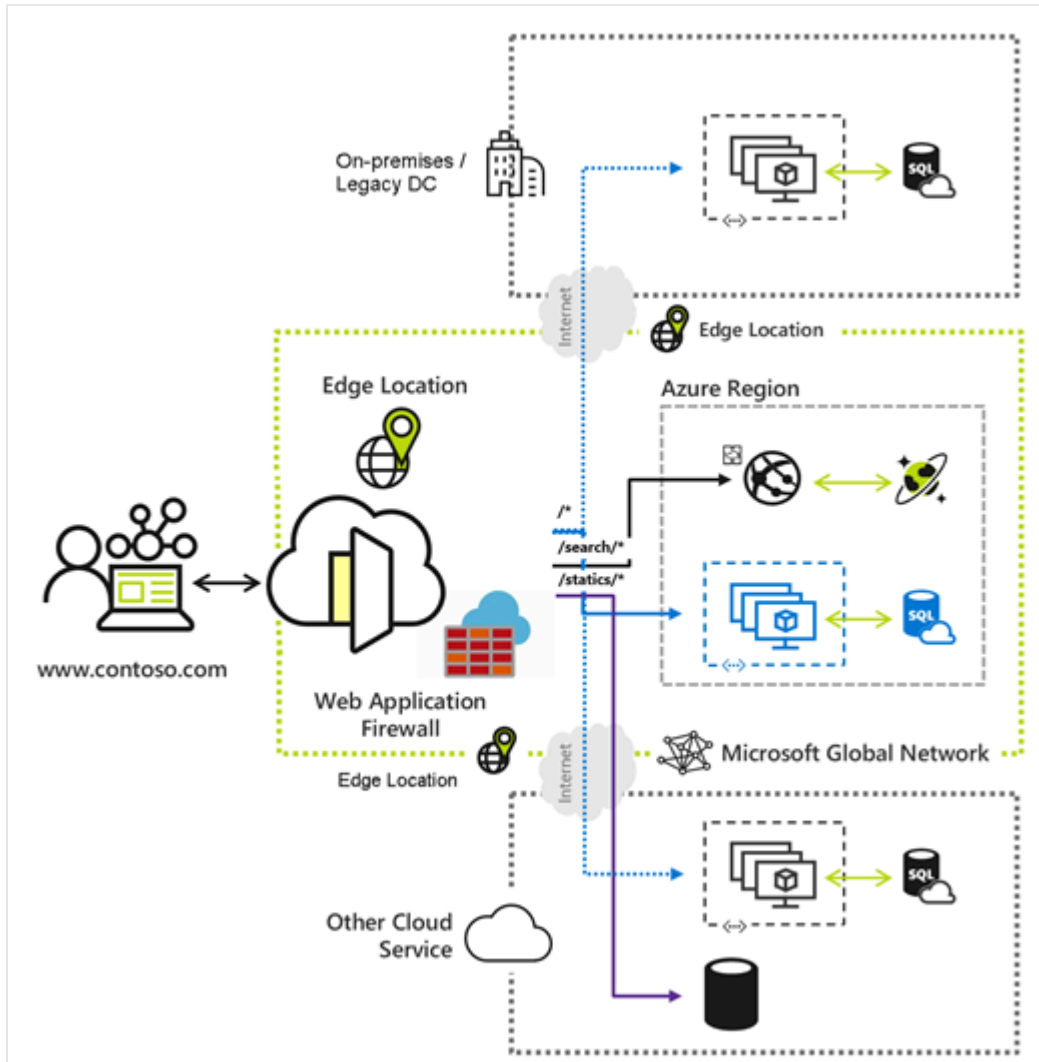
[Azure Application Gateway](#) is a web traffic load balancer that enables you to manage traffic to your web applications. It's an Application Delivery Controller (ADC) as a service, offering various layer 7 load-balancing capabilities for your applications.

The following diagram shows url path-based routing with Application Gateway.



Azure Front Door

Azure Front Door enables you to define, manage, and monitor the global routing for your web traffic by optimizing for best performance and instant global failover for high availability. With Front Door, you can transform your global (multi-region) consumer and enterprise applications, APIs, and content that reach a global audience with Azure.



Hybrid connectivity

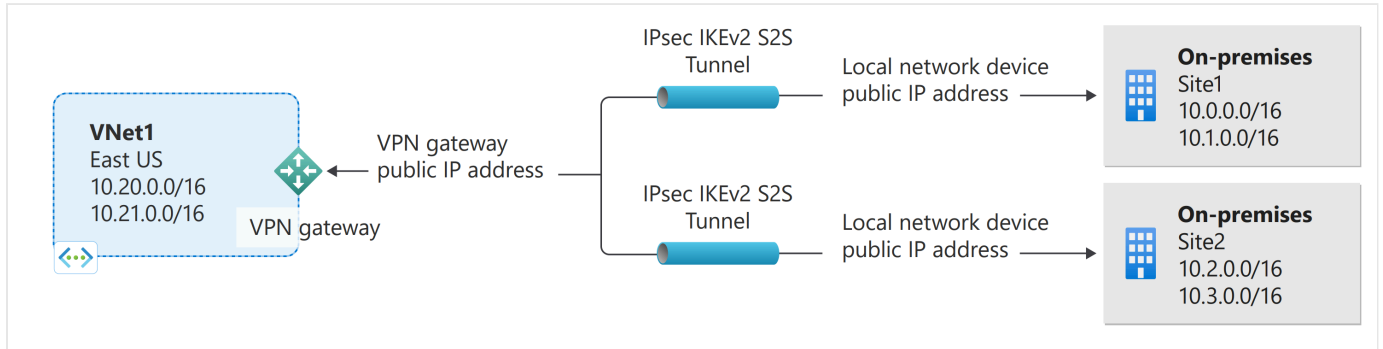
This section describes network connectivity services that provide a secure communication between your on-premises network and Azure - VPN Gateway, ExpressRoute, Virtual WAN, and Peering Service.

VPN Gateway

VPN Gateway helps you create encrypted cross-premises connections to your virtual network from on-premises locations, or create encrypted connections between VNets. There are different configurations available for VPN Gateway connections. Some of the main features include:

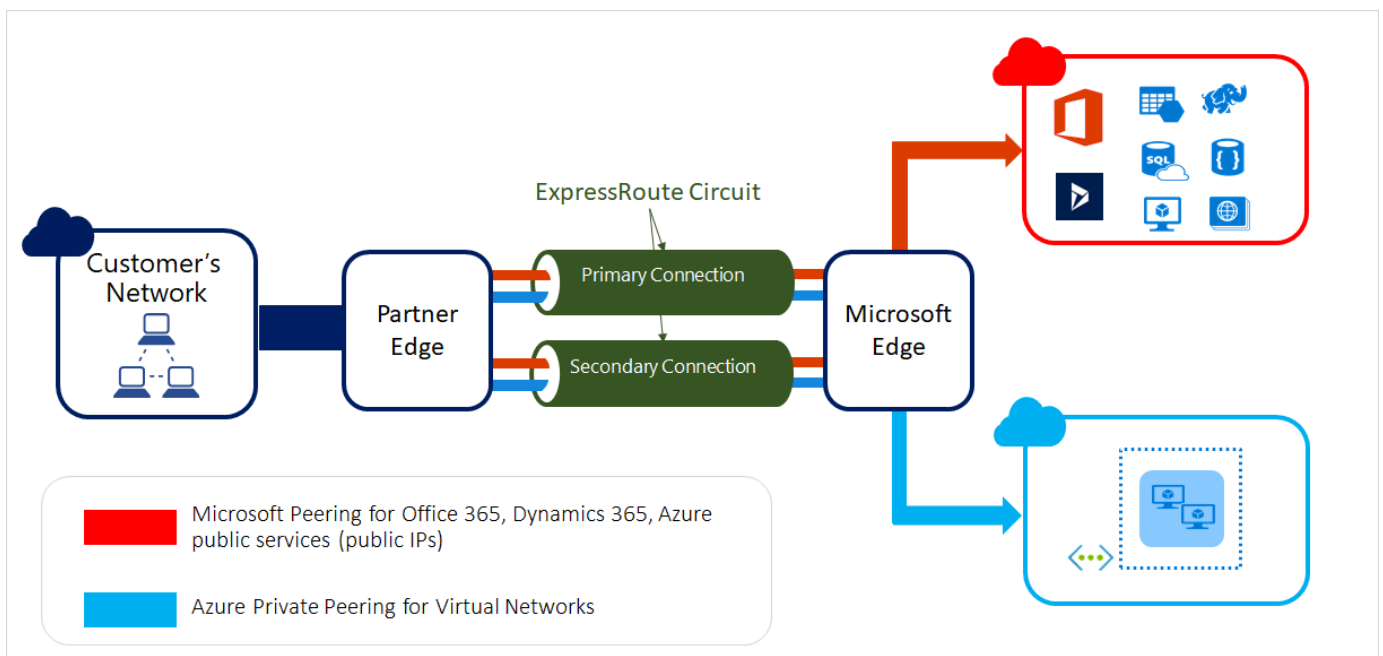
- Site-to-site VPN connectivity
- Point-to-site VPN connectivity
- VNet-to-VNet VPN connectivity

The following diagram illustrates multiple site-to-site VPN connections to the same virtual network. To view more connection diagrams, see [VPN Gateway - design](#).



ExpressRoute

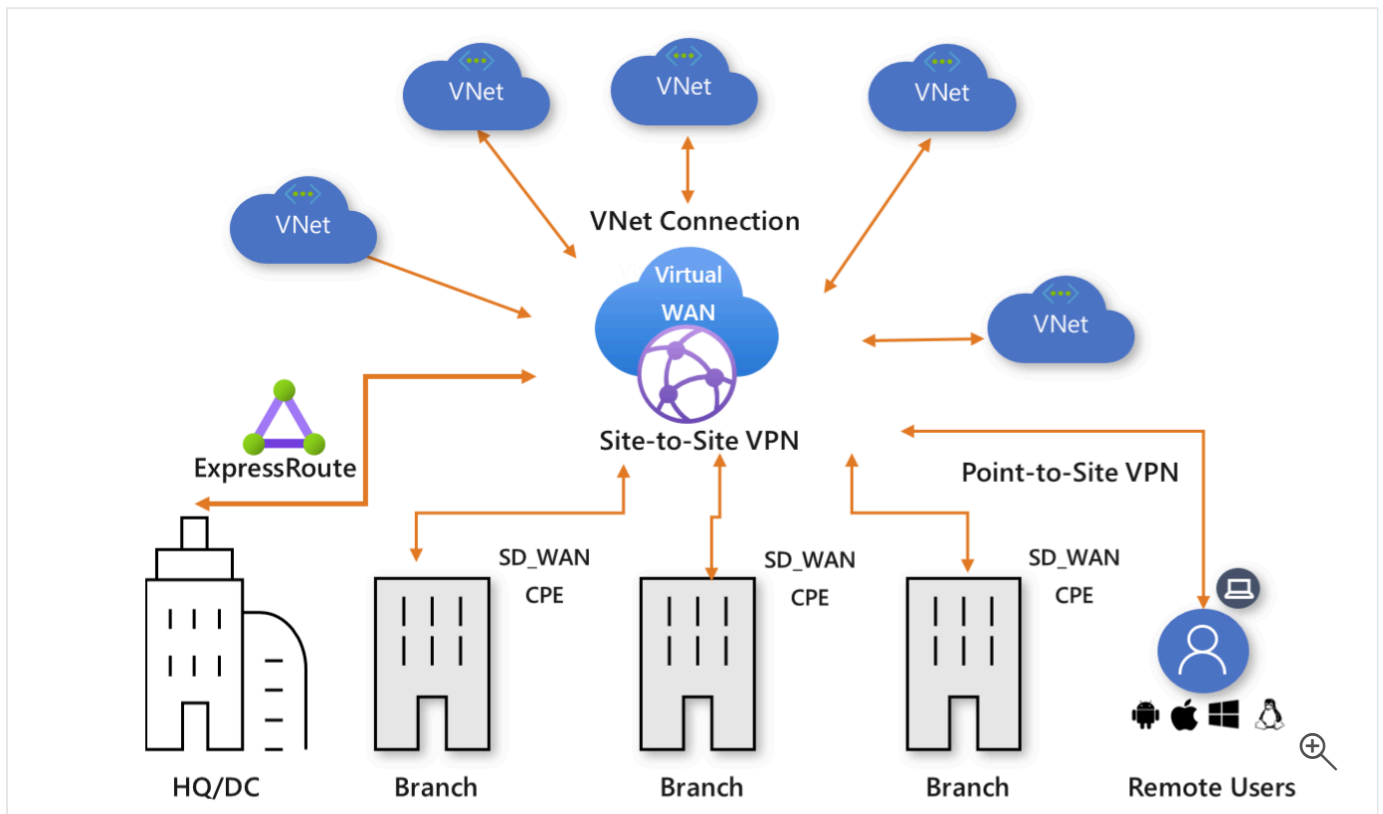
[ExpressRoute](#) enables you to extend your on-premises networks into the Microsoft cloud over a private connection facilitated by a connectivity provider. This connection is private. Traffic doesn't go over the internet. With ExpressRoute, you can establish connections to Microsoft cloud services, such as Microsoft Azure, Microsoft 365, and Dynamics 365.



Virtual WAN

[Azure Virtual WAN](#) is a networking service that brings many networking, security, and routing functionalities together to provide a single operational interface. Connectivity to Azure VNets is established by using virtual network connections. Some of the main features include:

- Branch connectivity (via connectivity automation from Virtual WAN Partner devices such as SD-WAN or VPN CPE)
- Site-to-site VPN connectivity
- Remote user VPN connectivity (point-to-site)
- Private connectivity (ExpressRoute)
- Intra-cloud connectivity (transitive connectivity for virtual networks)
- VPN ExpressRoute inter-connectivity
- Routing, Azure Firewall, and encryption for private connectivity



Peering Service

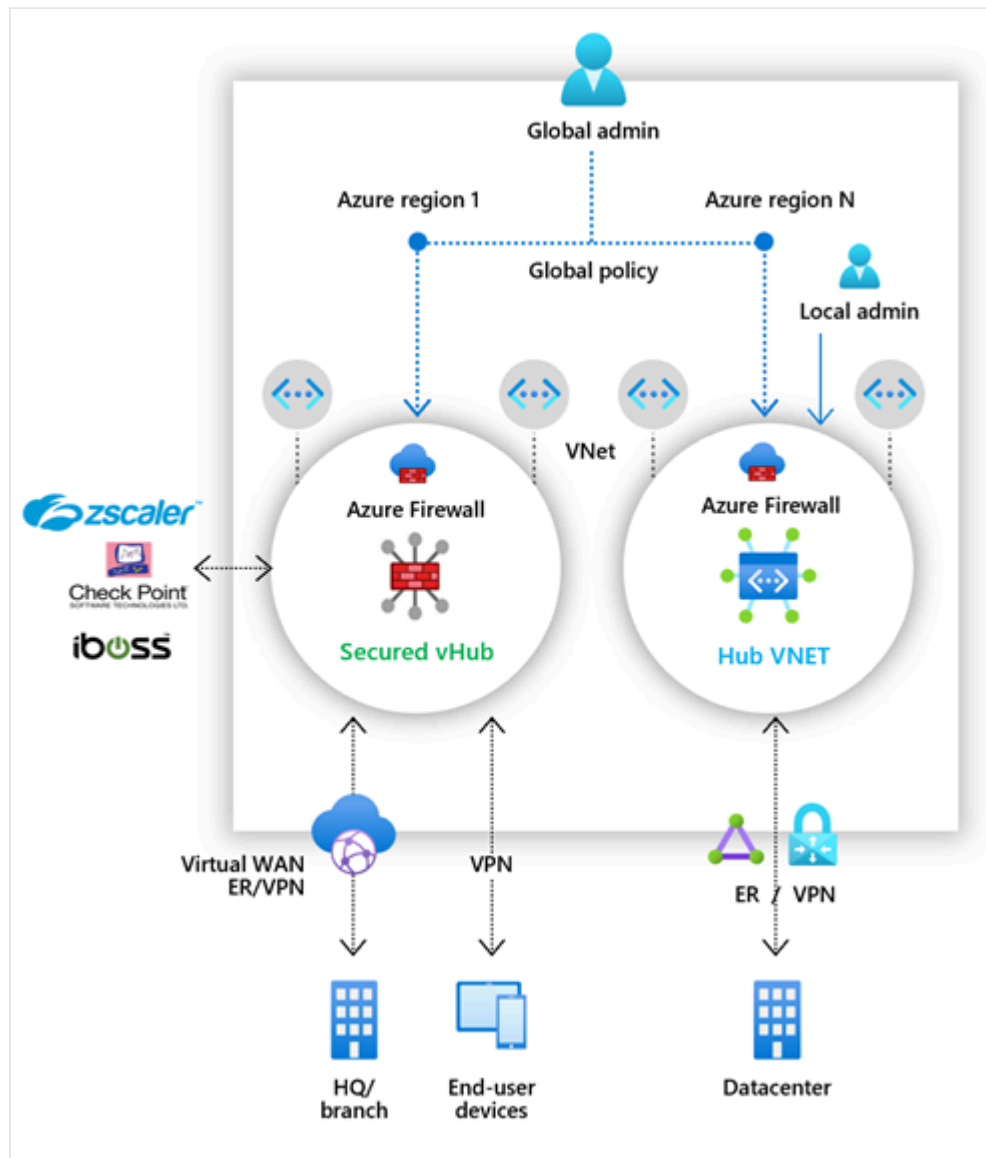
[Azure Peering Service](#) enhances customer connectivity to Microsoft cloud services such as Microsoft 365, Dynamics 365, software as a service (SaaS) services, Azure, or any Microsoft services accessible via the public internet.

Network security

This section describes networking services in Azure that protect and monitor your network resources - Firewall Manager, Firewall, Web Application Firewall, and DDoS Protection.

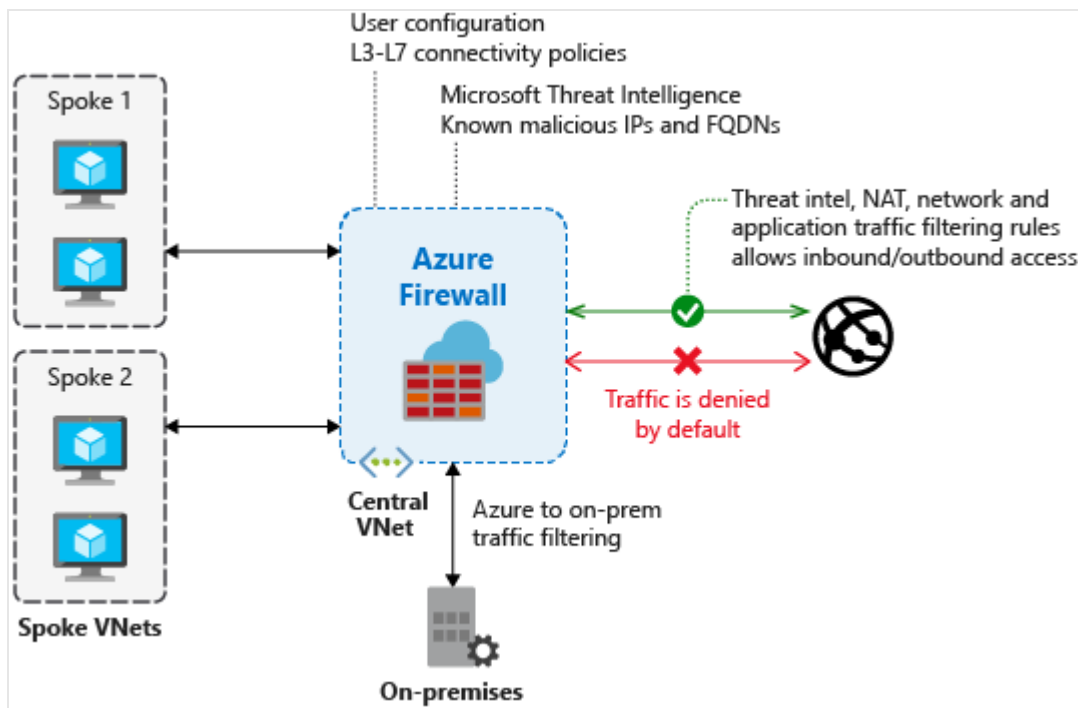
Firewall Manager

Azure Firewall Manager is a security management service that provides central security policy and routing management for cloud based security perimeters. Firewall manager can provide security management for two different types of network architecture: secure virtual hub and hub virtual network. With Azure Firewall Manager, you can deploy multiple Azure Firewall instances across Azure regions and subscriptions, implement DDoS protection plans, manage web application firewall policies, and integrate with partner security-as-a-service for enhanced security.



Azure Firewall

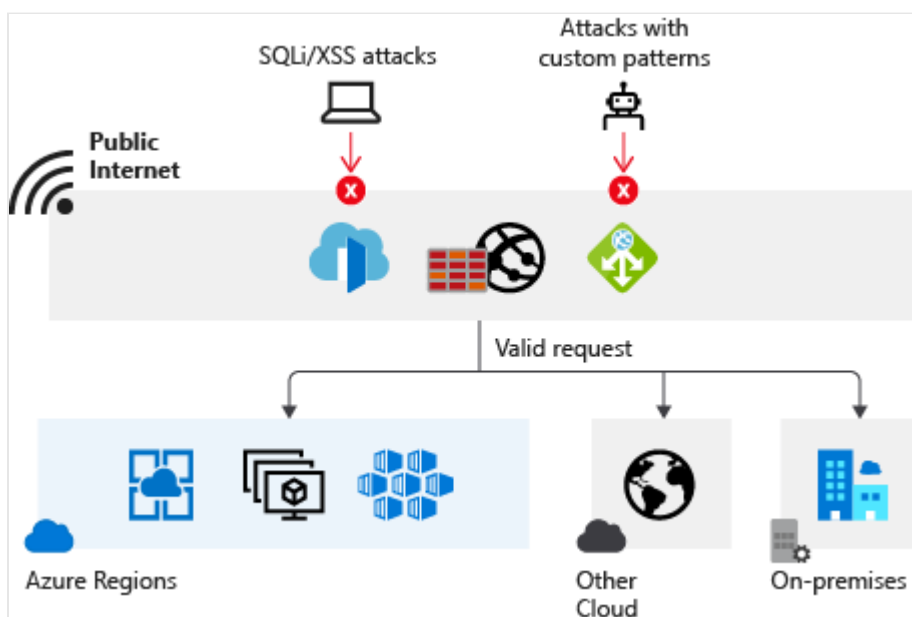
Azure Firewall is a managed, cloud-based network security service that protects your Azure Virtual Network resources. Using Azure Firewall, you can centrally create, enforce, and log application and network connectivity policies across subscriptions and virtual networks. Azure Firewall uses a static public IP address for your virtual network resources allowing outside firewalls to identify traffic originating from your virtual network.



Web Application Firewall

[Azure Web Application Firewall \(WAF\)](#) provides protection to your web applications from common web exploits and vulnerabilities such as SQL injection, and cross site scripting. Azure WAF provides out of box protection from OWASP top 10 vulnerabilities via managed rules. Additionally customers can also configure custom rules, which are customer managed rules to provide extra protection based on source IP range, and request attributes such as headers, cookies, form data fields, or query string parameters.

Customers can choose to deploy [Azure WAF with Application Gateway](#), which provides regional protection to entities in public and private address space. Customers can also choose to deploy [Azure WAF with Front Door](#) which provides protection at the network edge to public endpoints.

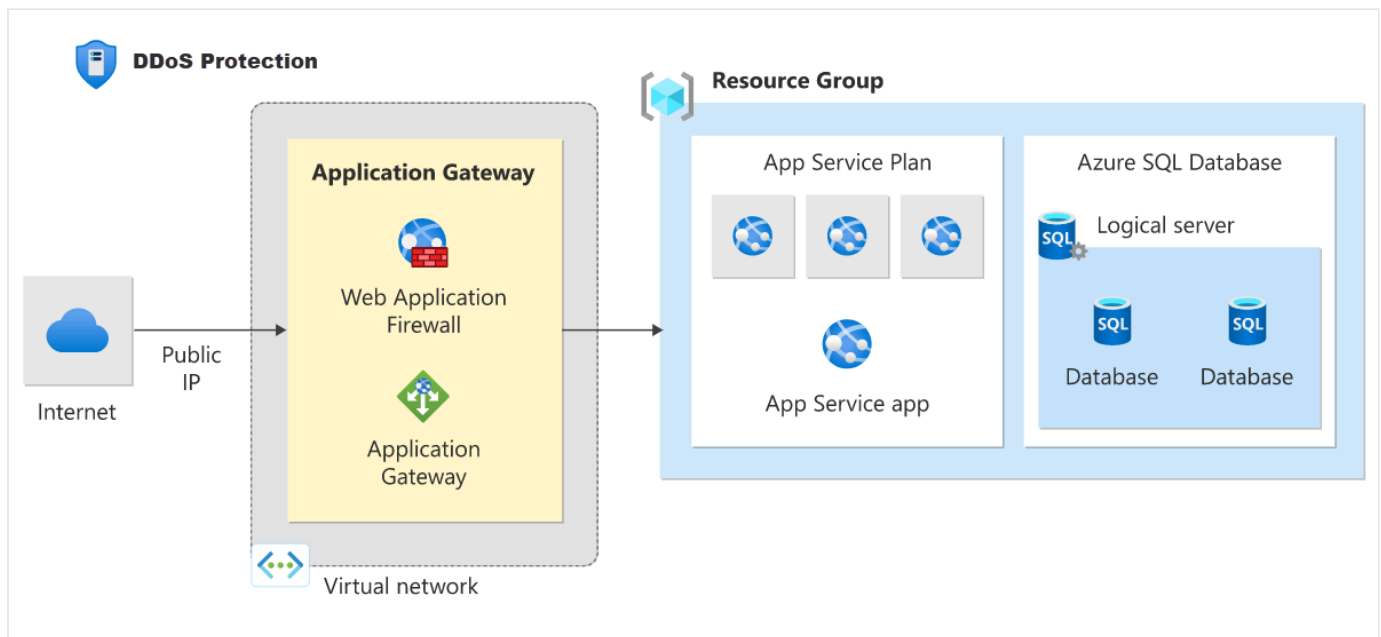


DDoS Protection

[Azure DDoS Protection](#) provides countermeasures against the most sophisticated DDoS threats. The service provides enhanced DDoS mitigation capabilities for your application and resources deployed in your virtual networks. Additionally, customers using Azure DDoS Protection have access to DDoS Rapid Response support to engage DDoS experts during an active attack.

Azure DDoS Protection consists of two tiers:

- [DDoS Network Protection](#), combined with application design best practices, provides enhanced DDoS mitigation features to defend against DDoS attacks. It's automatically tuned to help protect your specific Azure resources in a virtual network.
- [DDoS IP Protection](#) is a pay-per-protected IP model. DDoS IP Protection contains the same core engineering features as DDoS Network Protection, but differs in the following value-added services: DDoS rapid response support, cost protection, and discounts on WAF.



Container network security

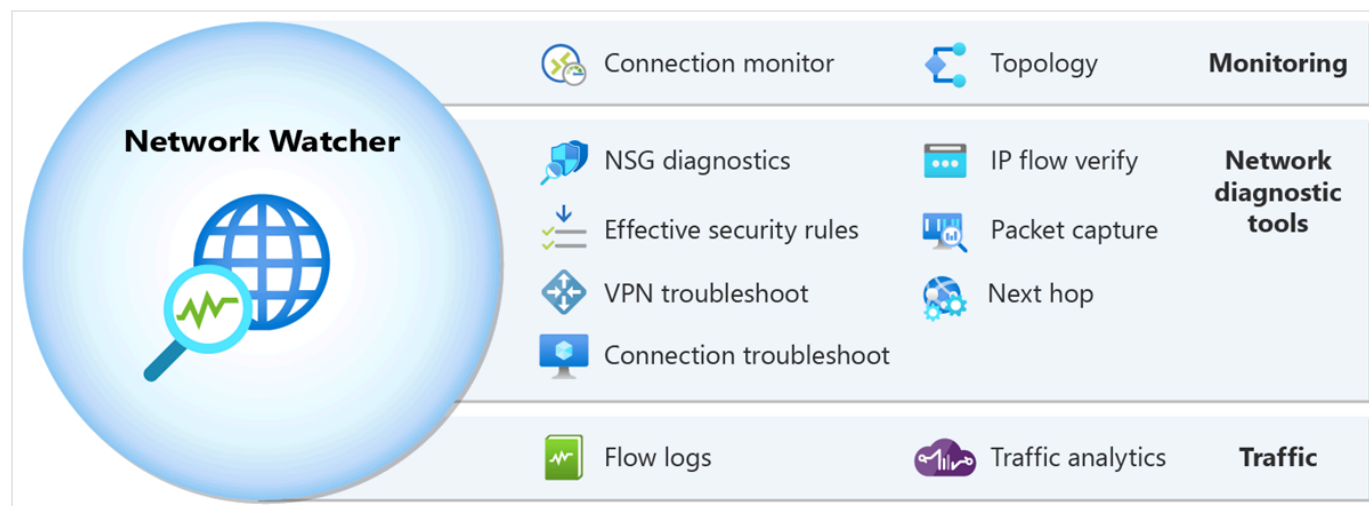
Container network security is part of [Advanced Container Networking Services](#). Advanced Container Networking Services provides enhanced control over AKS network security. With features like fully qualified domain name (FQDN) filtering, clusters using Azure CNI Powered by Cilium can implement FQDN-based network policies to achieve a Zero Trust security architecture in AKS.

Network Management and monitoring

This section describes network management and monitoring services in Azure - Network Watcher, Azure Monitor, and Azure Virtual Network Manager.

Azure Network Watcher

[Azure Network Watcher](#) provides tools to monitor, diagnose, view metrics, and enable or disable logs for resources in an Azure virtual network.



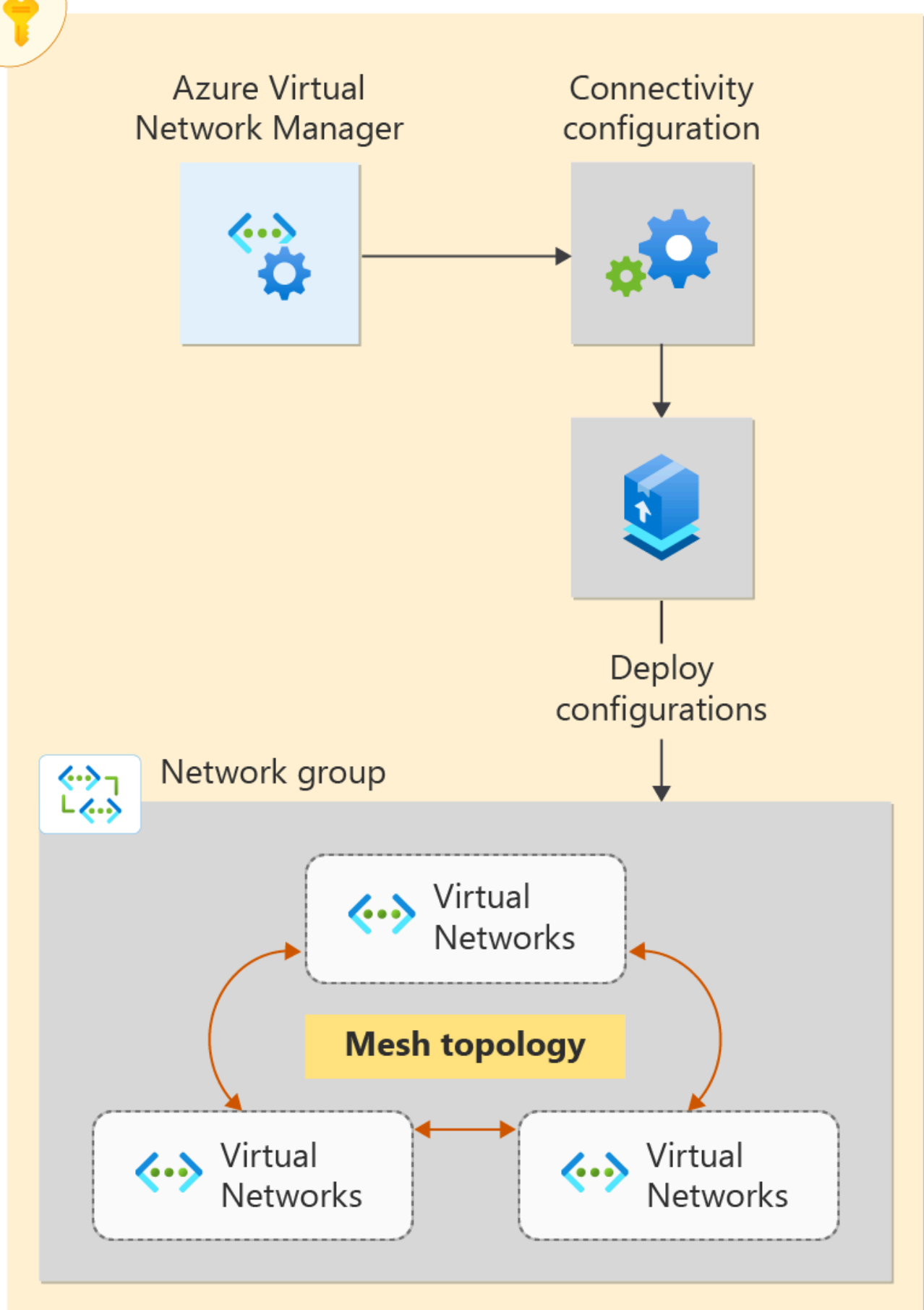
Azure Monitor

[Azure Monitor](#) maximizes the availability and performance of your applications by delivering a comprehensive solution for collecting, analyzing, and acting on telemetry from your cloud and on-premises environments. It helps you understand how your applications are performing and proactively identifies issues affecting them and the resources they depend on.

Azure Virtual Network Manager

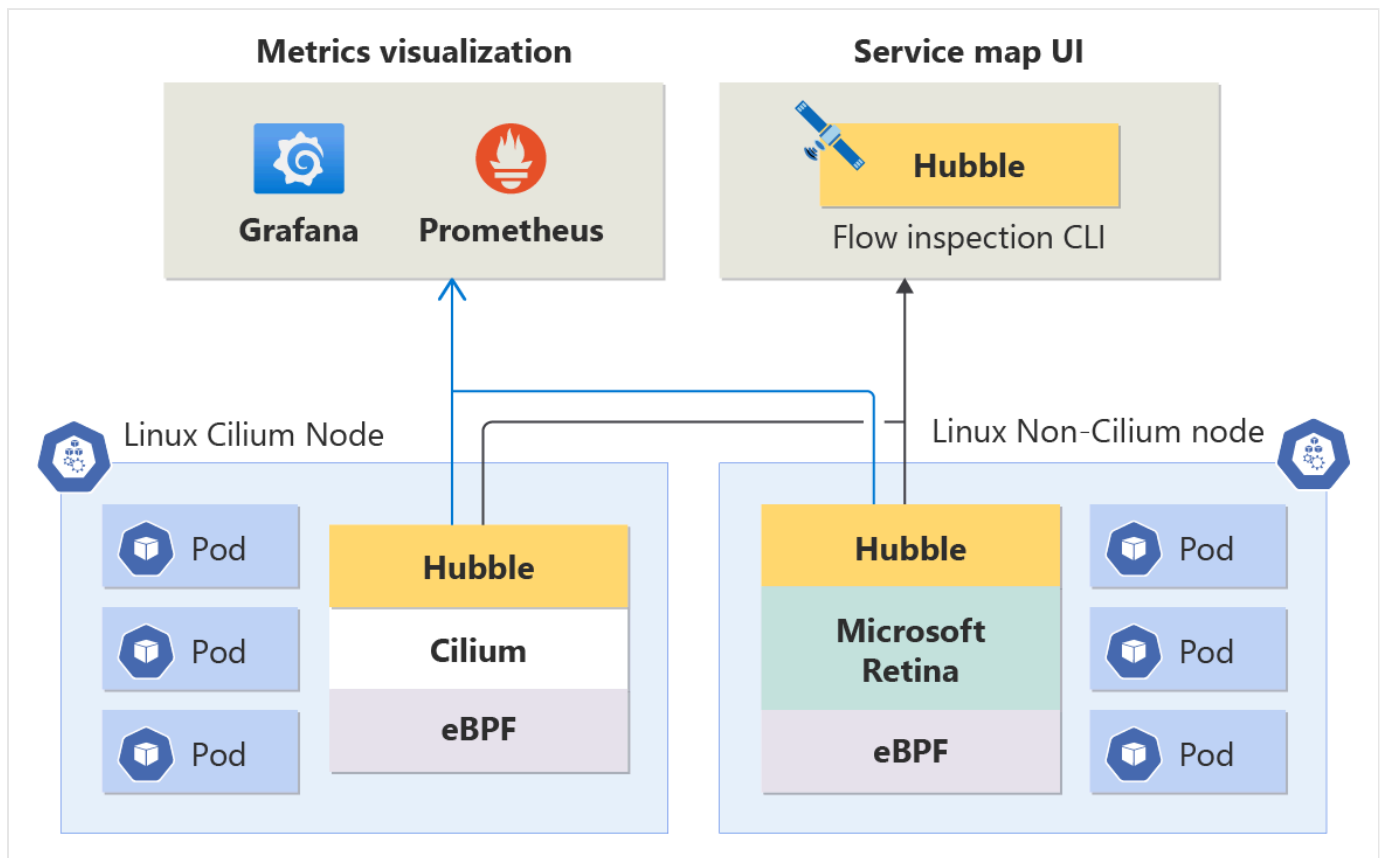
[Azure Virtual Network Manager](#) is a management service that enables you to group, configure, deploy, and manage virtual networks globally across subscriptions. With Virtual Network Manager, you can define [network groups](#) to identify and logically segment your virtual networks. Then you can determine the [connectivity](#) and [security configurations](#) you want and apply them across all the selected virtual networks in network groups at once.

Subscription-level Scope



Container network observability

Container network observability is part of [Advanced Container Networking Services](#). Advanced Container Networking Services uses Hubble's control plane to provide comprehensive visibility into AKS networking and performance. It offers real-time, detailed insights across node-level, pod-level, TCP, and DNS metrics, ensuring thorough monitoring of your network infrastructure.



Next steps

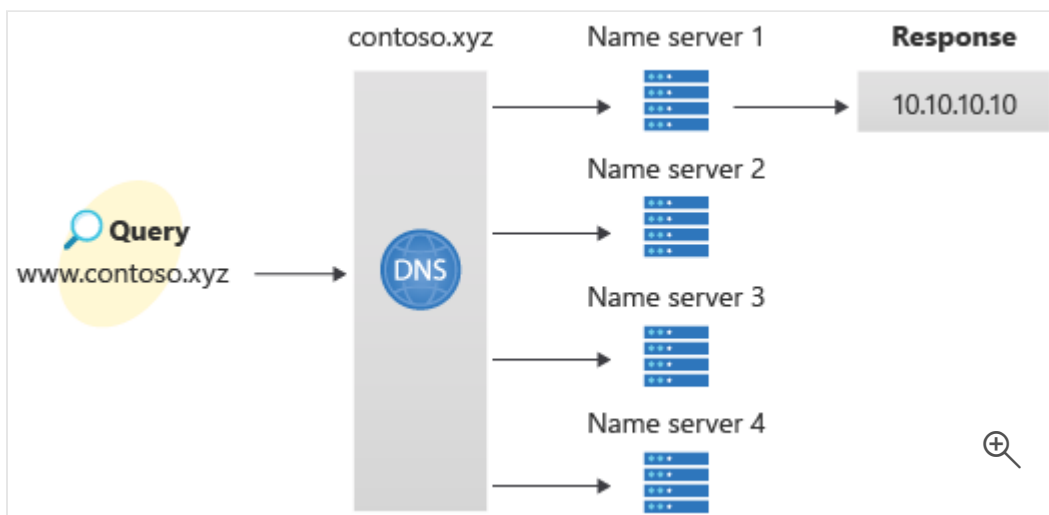
- Create your first virtual network, and connect a few virtual machines to it, by completing the steps in the [Create your first virtual network](#) article.
- Connect your computer to a virtual network by completing the steps in the [Configure a point-to-site connection](#) article.
- Load balance Internet traffic to public servers by completing the steps in the [Create an Internet-facing load balancer](#) article.

Last updated on 04/23/2026

Quickstart: Create an Azure DNS zone and record using the Azure portal

You can configure Azure DNS to resolve host names in your public domain. For example, if you purchased the *contoso.xyz* domain name from a domain name registrar, you can configure Azure DNS to host the *contoso.xyz* domain and resolve *www.contoso.xyz* to the IP address of your web server or web app.

In this quickstart, you create a test domain, and then create an address record to resolve *www* to the IP address *10.10.10.10*.



i Important

The names and IP addresses in this quickstart are examples that do not represent real-world scenarios. The private IP address *10.10.10.10* is used here with a public DNS zone for testing purposes.

You can also perform these steps using [Azure PowerShell](#) or the cross-platform [Azure CLI](#).

If you don't have an Azure subscription, create a [free account](#) before you begin.

For all portal steps, sign in to the [Azure portal](#).

Prerequisites

An Azure account with an active subscription is required. [Create an account for free](#).

Sign in to the Azure portal

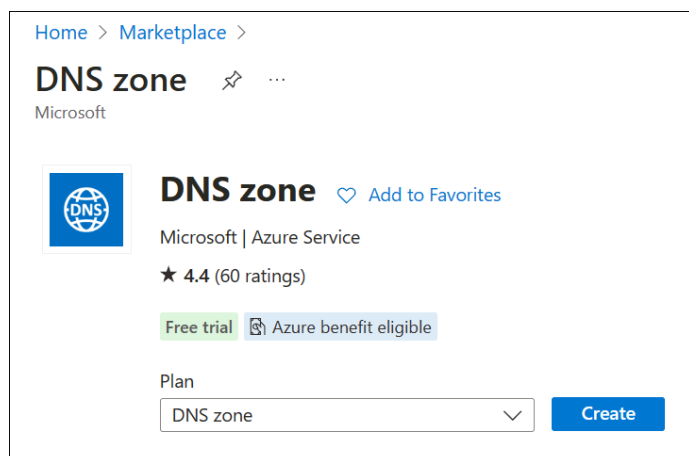
Sign in to the [Azure portal](#) with your Azure account.

Create a DNS zone

A DNS zone contains the DNS entries for a domain. To start hosting your domain in Azure DNS, create a DNS zone for that domain name.

To create the DNS zone:

1. At the upper left, select **Create a resource**, enter **DNS zone** into **Search services and marketplace** and then select **DNS zone**.
2. On the **DNS zone** page, select **Create**.



3. On the **Create DNS zone** page, type or select the following values:
 - **Resource group:** Select **Create new**, enter *MyResourceGroup*, and select **OK**. The resource group name must be unique within the Azure subscription.
 - **Name:** Type *contoso.xyz* for this quickstart example. The DNS zone name can be any value that isn't already configured on the Azure DNS servers. A real-world value would be a domain that you bought from a domain name registrar.
 - **Resource group location:** Select a location for the new resource group. In this example, the location selected is **West US**.
4. Select **Review + create** and then select **Create**.

[Home](#) > [Marketplace](#) > [DNS zone](#) >

Create DNS zone

Basics Tags Review + create

A DNS zone is used to host the DNS records for a particular domain. For example, the domain 'contoso.com' may contain a number of DNS records such as 'mail.contoso.com' (for a mail server) and 'www.contoso.com' (for a web site). Azure DNS allows you to host your DNS zone and manage your DNS records, and provides name servers that will respond to DNS queries from end users with the DNS records that you create. [Learn more.](#)

Project details

Subscription *

Resource group * [Create new](#)

Instance details

☐ This zone is a child of an existing zone already hosted in Azure DNS ⓘ

Name * ✓

Resource group location ⓘ

It may take a minute to create the zone.

Create a DNS record

Next, DNS records are created for your domain inside the DNS zone. A new address record, known as an 'A' record, is created to resolve a host name to an IPv4 address.

To create an 'A' record:

1. In the Azure portal, under **Resources**, open the **contoso.xyz** DNS zone.
2. At the top of the **contoso.xyz** DNS zone page, select **Record sets** and then select **+ Add**. The **Add a record set** window appears.
3. In the **Add a record set** window, enter or select the following values:
 - **Name:** Type *www*. This record name is the host name that you want to resolve to the specified IP address.
 - **Type:** Select **A**. 'A' records are the most common, but there are other record types for mail servers ('MX'), IP v6 addresses ('AAAA'), and so on.
 - **Alias record set:** Choose **No**.
 - **TTL:** Enter **1**. *Time-to-live* of the DNS request specifies how long DNS servers and clients can cache a response.

- **TTL unit:** Select **Hours**. The time unit for the **TTL** entry is specified here.
- **IP address:** For this quickstart example, enter **10.10.10.10**. This value is the IP address that the record name resolves to. In a real-world scenario, you would enter the public IP address for your web server.

4. Select **Add** to create the A record.

Since this quickstart is just for quick testing purposes, there's no need to configure the Azure DNS name servers at a domain name registrar. In a real production domain, you must enable users on the Internet to resolve the host name and connect to your web server or app. To accomplish this task, visit your domain name registrar and replace the name server records with the Azure DNS name servers. For more information, see [Tutorial: Host your domain in Azure DNS](#).

Test the name resolution

Now that you have a test DNS zone with a test 'A' record, you can test the name resolution.

To test DNS name resolution:

1. On the **contoso.xyz | Recordsets** page, copy one of the name server names from the name server list. For example: ns1-37.azure-dns.com.

Home > contoso.xyz

contoso.xyz | Recordsets ☆ ...

DNS zone

Search

+ Add Refresh Delete

A record set is a collection of records in a zone that have the same name and are the same type. Record Sets will be automatically fetched in batches of 100 as you scroll through the existing record sets. [Learn more](#)

Search

Fetched 3 record set(s).

Name	Type	TTL	Value
@	NS	172800	ns1-37.azure-dns.com. ns2-37.azure-dns.net. ns3-37.azure-dns.org. ns4-37.azure-dns.info.
@	SOA	3600	Email: azuredns-hostmaster.microsoft.com Host: ns1-37.azure-dns.com. Refresh: 3600 Retry: 300 Expire: 2419200 Minimum TTL: 300 Serial number: 1
www	A	3600	10.10.10.10

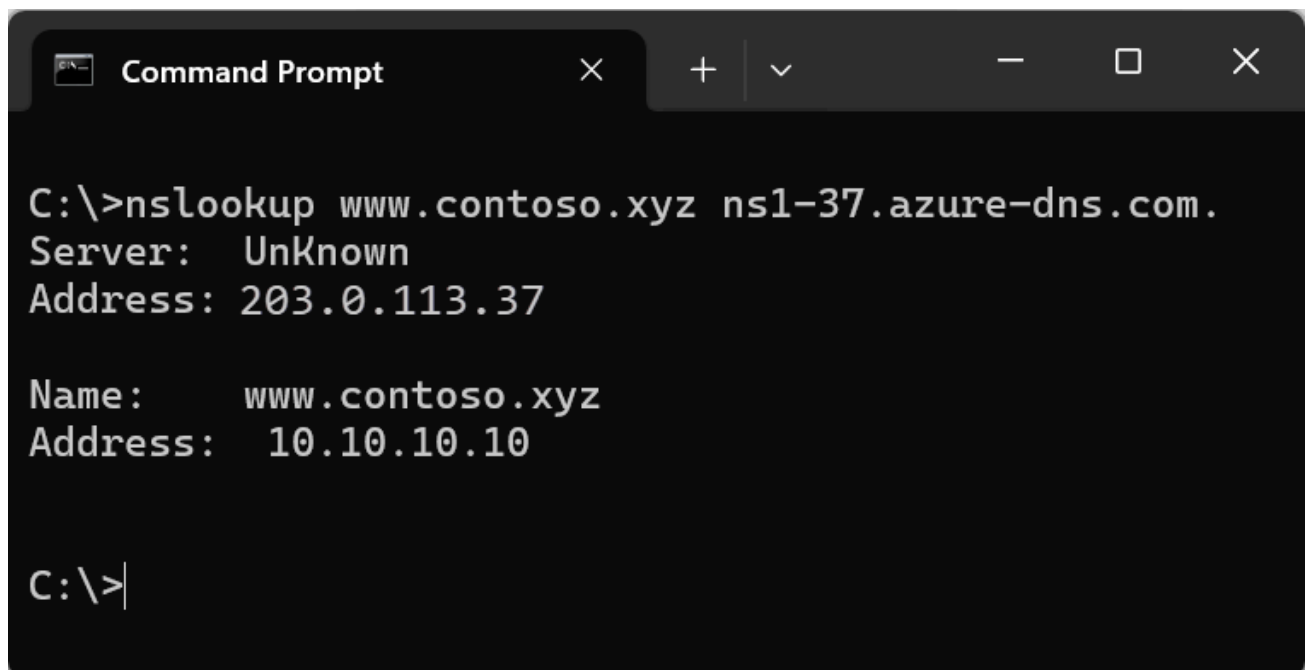
2. Open a command prompt, and run the following command:

```
nslookup www.contoso.xyz <name server name>
```

For example:

```
nslookup www.contoso.xyz ns1-37.azure-dns.com.
```

See the following example:



```
Command Prompt
C:\>nslookup www.contoso.xyz ns1-37.azure-dns.com.
Server:      UnKnown
Address: 203.0.113.37

Name:   www.contoso.xyz
Address: 10.10.10.10

C:\>
```

The host name **www.contoso.xyz** resolves to **10.10.10.10**, just as you configured it. This result verifies that name resolution is working correctly.

Clean up resources

When you no longer need the resources you created in this quickstart, remove them by deleting the **MyResourceGroup** resource group. Open the **MyResourceGroup** resource group, and select **Delete resource group**.

Next steps

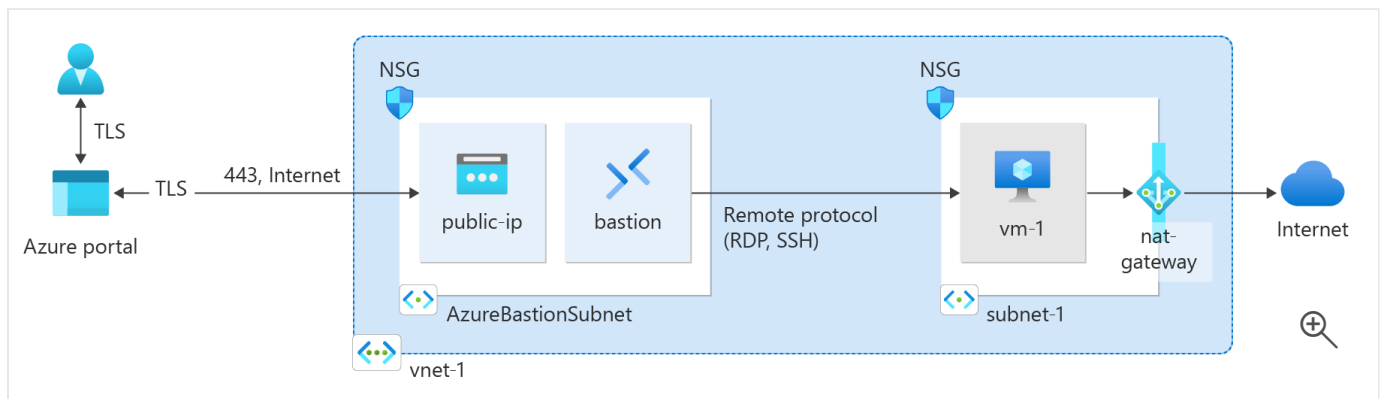
Create DNS records for a web app in a custom domain

Last updated on 06/11/2024

Quickstart: Create a Standard NAT gateway

In this quickstart, learn how to create a network address translation (NAT) gateway for the Standard SKU of Azure NAT Gateway by using the Azure portal, Azure PowerShell, or the Azure CLI. The Azure NAT Gateway service provides scalable outbound connectivity for virtual machines in Azure.

The following diagram shows the resources that you'll create in this quickstart.



Prerequisites

Portal

- An Azure account with an active subscription. [Create an account for free](#).

Create a resource group

Portal

1. In the search box at the top of the portal, enter **Resource groups**. Select **Resource groups** in the search results.
2. Select **+ Create**.
3. In **Create a resource group**, enter or select the following values:

[Expand table](#)

Setting	Value
Project details	
Subscription	Select your Azure subscription.
Resource group	Enter test-rg .
Resource details	
Region	Select (US) East US 2 .

4. Select **Review + create**.

5. Select **Create**.

Create a virtual network

Portal

The following procedure creates a virtual network with a resource subnet:

1. In the portal, search for and select **Virtual networks**.
2. On the **Virtual networks** page, select **+ Create**.
3. On the **Basics** tab of **Create virtual network**, enter or select the following information:

 Expand table

Setting	Value
Project details	
Subscription	Select your subscription.
Resource group	Select test-rg .
Instance details	
Name	Enter vnet-1 .
Region	Select (US) East US 2 .

4. Select **Next** to proceed to the **Security** tab.

5. Select **Next** to proceed to the **IP Addresses** tab.
6. In the address space box in **Subnets**, select the **default** subnet.
7. In **Edit subnet**, enter or select the following information:

[Expand table](#)

Setting	Value
Subnet purpose	Leave Default .
Name	Enter subnet-1 .
IPv4	
IPv4 address range	Leave the default of 10.0.0.0/16 .
Starting address	Leave the default of 10.0.0.0 .
Size	Leave the default of /24 (256 addresses) .

8. Select **Save**.
9. Select **Review + create** at the bottom of the pane. When the virtual network passes validation, select **Create**.

Deploy Azure Bastion

Portal

Azure Bastion uses your browser to connect to virtual machines (VMs) in your virtual network over Secure Shell (SSH) or Remote Desktop Protocol (RDP) by using their private IP addresses. The VMs don't need public IP addresses, client software, or special configuration. For more information, see [What is Azure Bastion?](#).

ⓘ Note

Hourly pricing starts from the moment that Bastion is deployed, regardless of outbound data usage. For more information, see [Pricing](#) and [SKUs](#). If you're deploying Bastion as part of a tutorial or test, we recommend that you delete this resource after you finish using it.

1. In the search box at the top of the portal, enter **Bastion**. Select **Bastions** in the search results.
2. Select **+ Create**.
3. On the **Basics** tab of **Create a Bastion**, enter or select the following information:

[Expand table](#)

Setting	Value
Project details	
Subscription	Select your subscription.
Resource group	Select test-rg .
Instance details	
Name	Enter bastion .
Region	Select (US) East US 2 .
Tier	Select Developer .
Configure virtual networks	
Virtual network	Select vnet-1 .

ⓘ Note

The Developer SKU for Azure Bastion is free and doesn't require a dedicated Azure Bastion subnet. For more information, see [Quickstart: Deploy Azure Bastion - Developer SKU](#).

4. Select **Review + create**.
5. Select **Create**.

Create a virtual machine

Portal

The following procedure creates a Linux virtual machine with SSH key authentication:

- 1. In the search box at the top of the portal, enter **Virtual machines**. Select **Virtual machines** in the search results.
- 2. Select **+ Create**, and then select **Azure virtual machine**.
- 3. On the **Basics** tab of **Create a virtual machine**, enter or select the following values:

 Expand table

Setting	Value
Project details	
Subscription	Select your Azure subscription.
Resource group	Select test-rg .
Instance details	
Virtual machine name	Enter vm-1 .
Region	Select (US) East US 2 .
Availability options	Select No infrastructure redundancy required .
Security type	Select Standard .
Image	Select Ubuntu Server 24.04 LTS - x64 Gen2 .
Size	Choose a size or leave the default setting.
Administrator account	
Authentication type	Select SSH public key .
Username	Enter azureuser .
SSH public key source	Select Generate new key pair .
Key pair name	Enter vm-1_key .

- 4. Select the **Networking** tab, or select **Next: Disks > Next: Networking**.
- 5. Select the following values:

 Expand table

Setting	Value
Network interface	

Setting	Value
Virtual network	Select vnet-1 .
Subnet	Select subnet-1 .
Public IP	Select None .

6. Select **Review + create**.

7. Review the settings, and then select **Create**.

8. When the **Generate new key pair** window opens, select **Download private key and create resource**. The key file is downloaded as **vm-1_key.pem**. Make sure you know where the .pem file is downloaded. You need the path to the key file to connect to the VM.

Create the NAT gateway

Portal

In this section, you create the NAT gateway resource and associate it with the subnet of the virtual network that you created.

1. In the search box at the top of the portal, enter **NAT gateway**. Select **NAT gateways** in the search results.
2. Select **+ Create**.
3. In **Create network address translation (NAT) gateway**, enter or select this information on the **Basics** tab:

 **Expand table**

Setting	Value
Project details	
Subscription	Select your Azure subscription.
Resource group	Select test-rg .
Instance details	

Setting	Value
NAT gateway name	Enter nat-gateway .
Region	Select (US) East US 2 .
SKU	Select Standard .
Availability zone	Select No Zone .
TCP idle timeout (minutes)	Leave the default of 4 .

For information about availability zones and NAT gateway, see [Reliability in Azure NAT Gateway](#).

4. Select the **Outbound IP** tab, or select **Next: Outbound IP**.

5. Enter or select the following information:

 Expand table

Setting	Value
Public IP addresses	Select Create a new public IP address . In Name , enter public-ip-nat . Select OK .

6. Select the **Networking** tab, or select **Next: Networking**.

7. In **Virtual network**, select **vnet-1**.

8. In **Subnet name**, select the **subnet-1** checkbox.

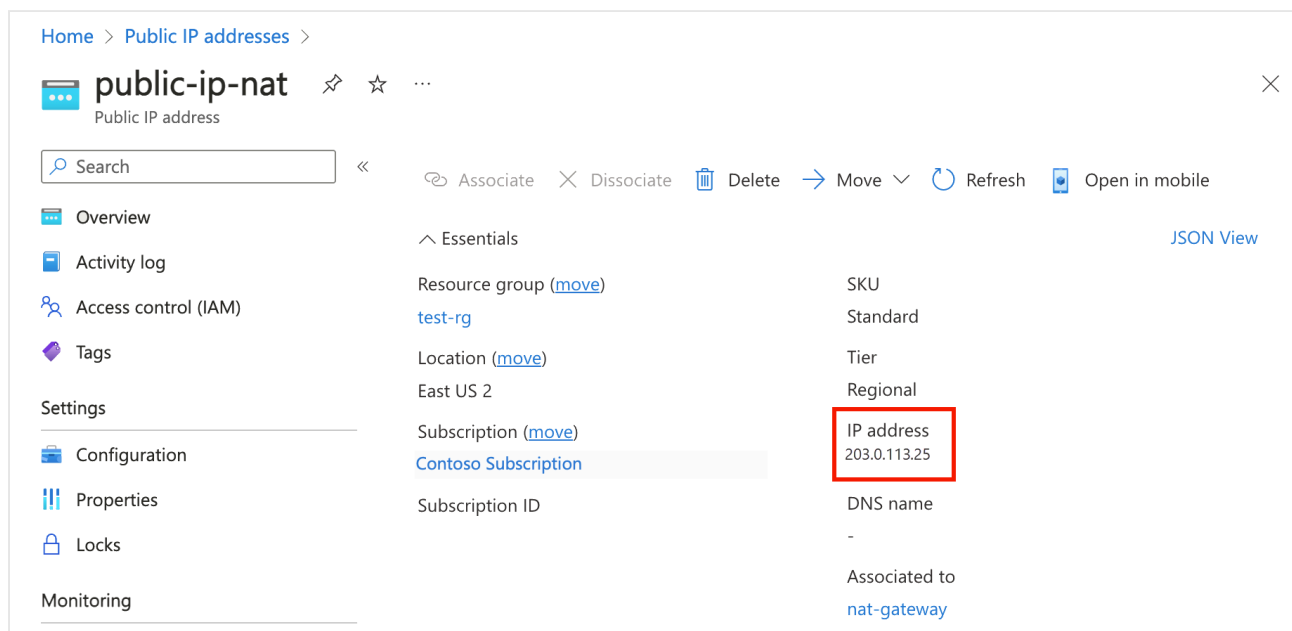
9. Select the **Review + create** tab, or select the **Review + create** button at the bottom of the pane.

10. Select **Create**.

Test the NAT gateway

To test the NAT gateway, you first discover its public IP. You then connect to the test virtual machine and verify the outbound connection through that public IP.

1. In the search box at the top of the portal, enter **Public IP**. Select **Public IP addresses** in the search results.
2. Select **public-ip-nat**.
3. Make note of the public IP address.



4. In the search box at the top of the portal, enter **Virtual machine**. Select **Virtual machines** in the search results.
5. Select **vm-1**.
6. On the **Overview** page, select **Connect**, and then select the **Bastion** tab.
7. Select **Use Bastion**.
8. Under **Authentication Type**, select **SSH Private Key from Local File**.
9. In **Username**, enter **azureuser**.
10. Select **Browse** and go to the **vm-1_key.pem** file downloaded during VM creation.
11. Select **Connect**.
12. In the Bash prompt, enter the following command:

Bash

```
curl ifconfig.me
```

13. Verify that the IP address returned by the command matches the public IP address of the NAT gateway.

Output

```
azureuser@vm-1:~$ curl ifconfig.me  
203.0.113.0.25
```

Clean up resources

Portal

When you finish using the resources that you created, you can delete the resource group and all its resources.

1. In the Azure portal, search for and select **Resource groups**.
2. On the **Resource groups** page, select the **test-rg** resource group.
3. On the **test-rg** page, select **Delete resource group**.
4. Enter **test-rg** in **Enter resource group name to confirm deletion**, and then select **Delete**.

Related content

- [Azure NAT Gateway overview](#)
- [NAT gateway resource](#)

Last updated on 05/01/2026